



U.S. ARMY JOINT MUNITIONS COMMAND

BUILDER Sustainment Guide

2023





BUILDER Sustainment Guide

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Forward

In September 2013, the Office of the Under Secretary of Defense for Acquisition, Technology and Logistics issued a policy memo mandating the use of the U.S. Army Corps of Engineers Engineer Research and Development Center (ERDC) Construction Engineering Research Laboratory's (CERL) Sustainment Management System (SMS), to "establish a DoD-wide facility condition assessment process [that] will contribute to a more credible DoD asset management program...and allow Department leadership to better target fiscal resources to those facilities most in need of investment."

The BUILDER Sustainment Management System (SMS) is a program that aggregates facility condition information to provide Senior Army Leaders with quality, reliable data to effectively invest limited resources while maintaining mission readiness. BUILDER program sustainment needs to be standardized to maintain consistency and reliability of the data; this contributes to command-wide credibility and defensibility of BUILDER moving forward. This Sustainment Guide outlines the resources necessary to maintain the BUILDER database for the foreseeable future and lays out the methods for systematically updating the existing data under various scenarios. It provides user training outlines and step-by-step instructions to keep the data current and quality control procedures to ensure the data meets Army standards.

Our mission, to "Provide the Joint Force with ready, reliable, lethal munitions at the speed of war sustaining global readiness," can be achieved most effectively if our aging infrastructure and facilities are sustained and modernized at a pace equal to today's technology and ever-expanding mission requirements. Sustaining BUILDER is essential to our mission by providing commanders with accurate facility inventory and condition data to make smart, informed investment decisions.

Provide Lethality that Wins!

/ s /

Corey Hotle

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1. Introduction

For years, the Department of Defense (DoD) has been plagued with insufficient funding necessary to support aging facilities and infrastructure. Sustainment, Restoration and Modernization (SRM) requirements have been historically underfunded, forcing the Services to accept risk to mission within their respective real property portfolios year after year.

Joint Munitions Command (JMC) has some of the oldest facilities in the U.S. Army's Organic Industrial Base (OIB) and relies on Government and Contractor personnel to operate and maintain the aging infrastructure. As such, JMC personnel must sustain accurate BUILDER data as a critical element in the Army's OIB modernization strategy, annual Facility Investment Planning, and installation readiness reporting to provide commanders the most accurate picture possible of the health of their facilities.

1.1 Purpose of the Guide

The purpose of the document is to provide guidance on how to implement a standardized approach to BUILDER data sustainment for JMC with alignment to the Deputy Chief of Staff (DCS), G-9, and the SMS Mission to "... assist DoD and U.S. Army leadership with information about real property asset condition, functionality, and remaining service life." According to the Army's Engineer Research and Development Center, Construction Engineering Research Laboratory (ERDC-CERL) and DCS, G-9, BUILDER provides the capability to:

- Implement objective, repeatable facility condition assessments
- Forecast asset degradation and identify the right time to invest to maintain facility/asset performance
- Enforce consistent condition and investment practices across the Enterprise
- Support development of a strategic approach to facility investments

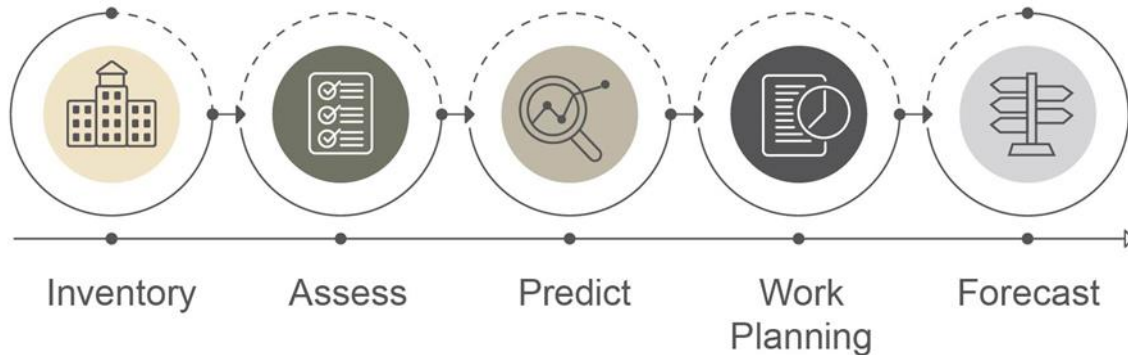


Figure 1: Overview of BUILDER Activities

The overarching assumption of this Guide is that JMC installation BUILDER data meets the current "Army BUILDER SMS Inventory and Assessment Guide" standards for quality and completeness. Any buildings that were excluded from the initial BUILDER implementation due to size (< 1,000 Square Feet (SF)), new construction, ongoing renovations or funding constraints, should also be captured in BUILDER to properly represent the installation's full Real Property inventory and condition. Prior to implementing this Sustainment Guide, all BUILDER data shall meet the standards outlined in the [Assessment Guide] as directed by DCS G-9.

1.1.1 Definition of BUILDER Sustainment

JMC defines BUILDER Sustainment as a process for maintaining the quality and integrity of the SMS data through periodic asset inspections; adding, removing, and updating data from completed projects or planned maintenance; and applying a prioritized, risk-based approach to keeping data current in support of condition-based capital investment and facility modernization planning for the U.S. Army's OIB.

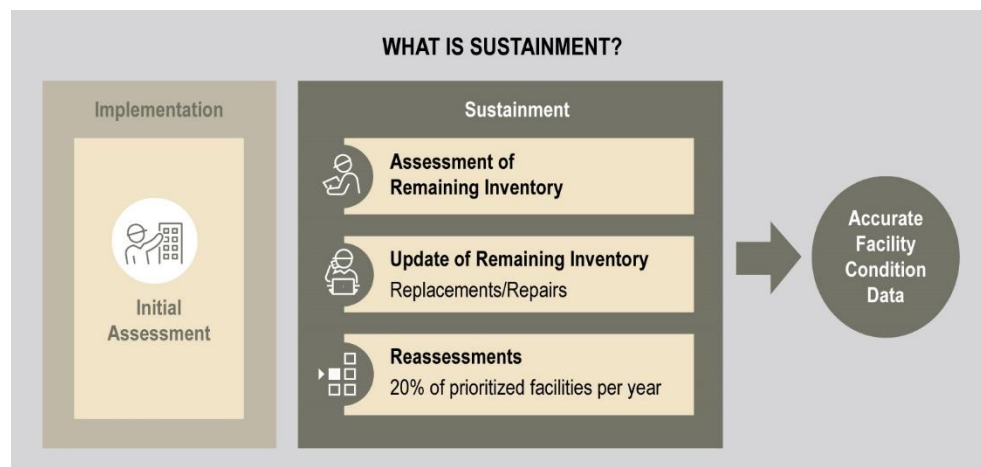


Figure 2: What is Sustainment

1.1.2 Current State of BUILDER

According to ERDC-CERL (Annex I to IMCOM OPORD 22-012), the initial BUILDER assessments established the baseline facility inventory and condition utilizing the Direct Condition Rating (DCR) criteria (Green +/-, Amber +/-, Red +/-) for all components to establish the Building Condition Index (BCI). BUILDER's algorithm continuously adjusts the asset's condition rating in the BCI by using a lifecycle degradation curve. This enables the data to stay current, assuming the component or asset "receives regular preventative maintenance and degrades according to average lifecycle curves."

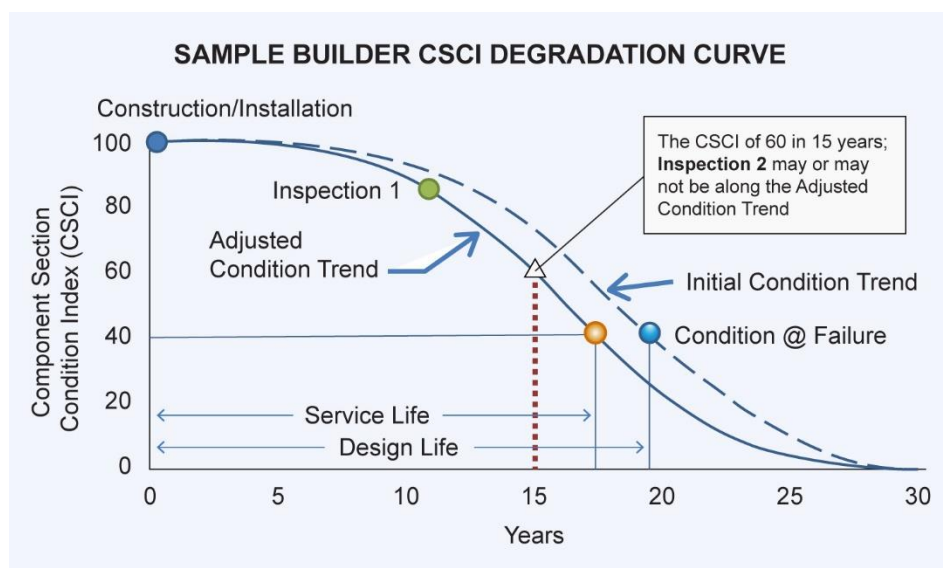


Figure 3: Sample BUILDER Component Degradation

BUILDER data must be sustained to ensure the accuracy of its metrics such as BCI, Facility Condition Index (FCI), etc., which JMC and Army Senior Leaders are using in master planning activities.

JMC BUILDER Data Sustainment Current Assumptions

- Initial BUILDER inventory and assessments have occurred at JMC installations
- Facilities that were not included in the initial implementation of BUILDER will be captured through a dedicated AMC/G4 assessment contract
- All BUILDER data meets current “Army BUILDER Inventory and Assessment Guide” standards for quality and completeness
- Regular preventative maintenance is occurring for assets at routine intervals to the greatest extent possible
- Asset degradation has occurred in conjunction with “normal” operations and maintenance parameters

1.2 Roadmap for JMC BUILDER Sustainment

This Guide provides a roadmap for JMC BUILDER Sustainment to drive ongoing data integrity. Whether the user has extensive BUILDER experience or is just beginning in their role as a BUILDER POC, this Guide will highlight the resources, training, and required annual activities to meet JMC’s BUILDER sustainment goals. Program Roles and Responsibilities are detailed in Appendix B.

1.2.1 Foundational Sustainment Activities

It is important to highlight the context of two foundational Sustainment activities and their source information. The first activity is to update BUILDER information for JMC buildings. The Army BUILDER Inventory and Assessment Guide is the official document to be followed when updating BUILDER. The second activity is prioritizing building reassessments for JMC. Guidance for this task is discussed primarily in this BUILDER Sustainment Guide.

It is understood that the reader can refer to both guides to support the mutual success of both inventory updates and prioritizing reassessments. All installations shall follow the “BUILDER Business Rules” as defined in Appendix E to identify the proper Real Property Asset (RPA) Type Code and other qualifying RP criteria for assets to be entered in or excluded from BUILDER.

1.2.2 Milestones for JMC BUILDER Sustainment

The JMC BUILDER Sustainment Timeline (Figure 4) depicts the key sustainment activities that must occur in Q1-Q3 of each fiscal year (FY) October-June. These BUILDER sustainment activities will support JMC, AMC, and the Army’s mandatory Real Property reporting to the Office of the Secretary of Defense (OSD) that occurs on 30 September each fiscal year.

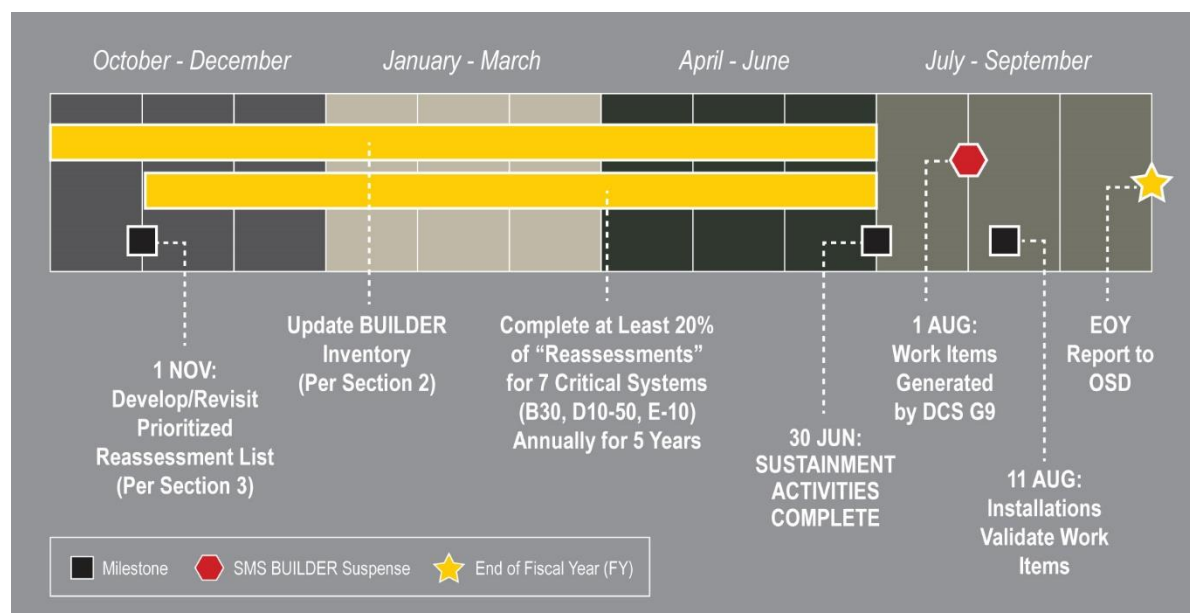


Figure 4: JMC BUILDER Sustainment Timeline

Each year, CERL will run the Annual Work Plan for the Army (DCS, G-9) on 1 August to produce the FCIs required in the annual OSD report. **Therefore, all data sustainment activities outlined in this Guide must be complete by the Installation POCs and validated by HQ JMC no later than the end of the third fiscal quarter (30 June) per DSC, G-9.**

Note: Any BUILDER update completed after 30 June may not be included in the G-9 annual work plan.

Section 2 provides guidance on conducting sustainment activities, including qualifying sustainment events that drive inventory updates. Real Property inventory should be updated in BUILDER when asset conditions change. This could be due to:

- New construction / completed repair projects
- Removal, replacement, or addition of assets / components
- Demolition

Section 3 of this Guide details critical steps to prioritization and sequencing of BUILDER data sustainment activities, to identify the most mission-critical facilities and systems for JMC that will require building-level reassessment. As depicted in Figure 4: JMC BUILDER Sustainment Timeline, it is recommended that development of this prioritized reassessment list should be completed by 1 November and reviewed annually in Q1 to validate what assessments have been completed and what facilities remain for the coming year.

Additional support material is contained in the Appendices, including Quality Control (QC) best practices, BUILDER Roles and Responsibilities, Training for BUILDER and the BUILDER Remote Entry Database (BRED) software, and how to download useful custom reports in BUILDER.

1.2.3 Resources

Several resources are available to assist Installation POCs with BUILDER.

The AMC G4 SMS Support desk (buildersmssupport@jacobs.com) can provide a wide variety of BUILDER services. These services include assisting the installations with collecting asset inventory and condition assessment data, creating custom reports or analyzing BUILDER data to assist in project funding justifications. The support desk can also answer technical questions regarding implementation of the Inventory and Assessment Guide, or this Sustainment Guide.

Additional support can also be found at ERDC-CERL SMS Support (SMSSupport@erdc.dren.mil) for technical issues with BUILDER.

Guidance documents:

- DCS, G-9 Installations, Real Property Asset Management Office, SMS BUILDER Sustainment Guidance
- HQ IMCOM OPORD 22-012 BUILDER Sustainment Plan (CUI), including all Annexes
- ERDC-CERL TN-06-1

2. Updating BUILDER Inventory

Once an installation's real property inventory is entered in BUILDER, the SMS algorithms immediately begin to degrade the assets over time, based on age and condition, essentially sustaining itself. The prioritization (filtering) process outlined in Section 2.1 provides step-by-step instructions for identifying JMC's critical facilities for reassessment within a five-year timeframe to meet DCS, G-9's intent for sustainment. However, there are several new or updated real property inventory "qualifying events" that represent the minimum data sustainment requirements that shall also be managed annually for each installation.

2.1 Qualifying sustainment events

Although BUILDER provides daily updates to the CI, certain "qualifying events"—as shown in Figure 5: Qualifying Events That May Dictate a Need for Updating BUILDER Inventory—may dictate a need for updating the BUILDER inventory that render the BCI inaccurate. For example, as new components are installed, the old component data should be removed from BUILDER and a new component section created in its place.

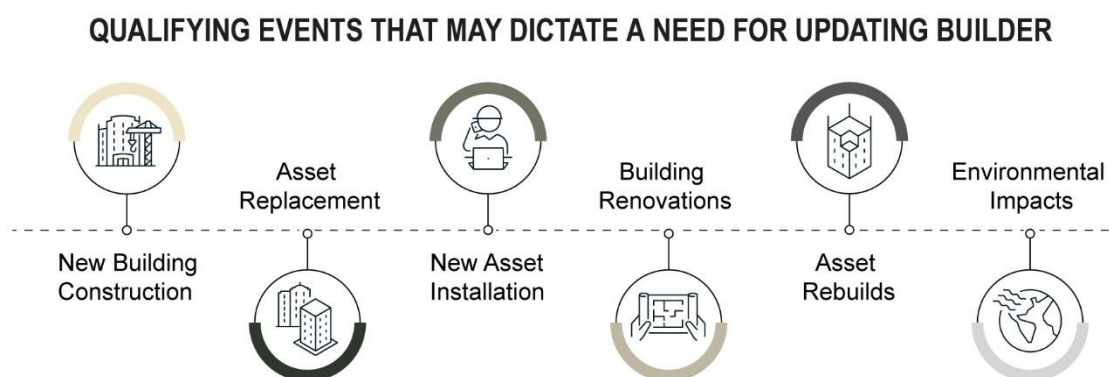


Figure 5: Qualifying Events That May Dictate a Need for Updating BUILDER Inventory

Similarly, new building construction or renovations must be added to BUILDER following the Government's acceptance of the work or when work items are completed. DCS, G-9 and CERL are developing a linkage between BUILDER and General Fund Enterprise Business System (GFEBS) to capture completed work, but until that system integration is deployed, updates to BUILDER must be accomplished manually.

It is important to become familiar with the BUILDER Inventory and Assessment Guide and review the U.S. Army BUILDER Reassessment Videos at <https://training.sms.erd.c.dren.mil> before adding or editing data in BUILDER or BRED. Choosing the correct system, component, and year installed is critical for BUILDER to establish the proper degradation curve and for CI to plan and budget for the future. An accurate inventory and assessment process takes time and cannot be rushed. Be aware that some components require "section details", photos, or specific comments. For more information on available training, please see Appendix C.

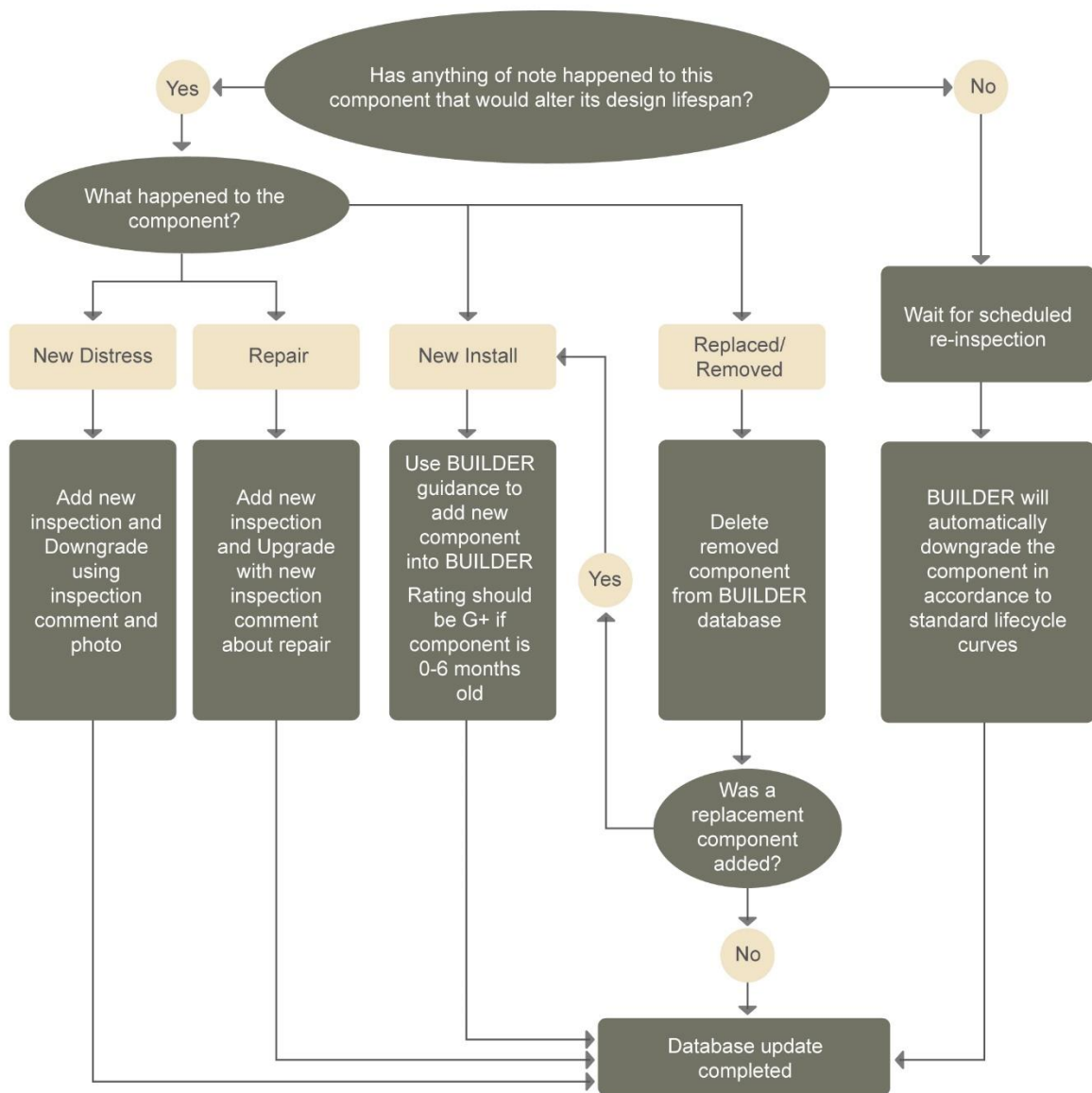
Drawings are enormously helpful during this process to keep track of spaces within the building, verify space configurations and size, and ensure that current room names or uses are consistent with the drawing. This is a good opportunity to update existing drawings as needed.

When working with existing data, do not assume all the data is accurate, complete, or all components have been collected. It is important to review all sections in the space/area/room during the reassessment. BRED will auto populate the Year Installed based on several factors; therefore, it is extremely important to review and correct this field when adding new sections.

It is important to think about complementary components within a system that may be missing or require an update. For example, assessment of a boiler should also include its associated hot water pump and piping, a multi-story building will have stairs and railings (and maybe an elevator), and a roof will have more than just the covering. It should include the support framing and may have a roof deck that are all assessed separately in BUILDER per the Inventory and Assessment Guide.

Figure 6 depicts CERL's decision tree regarding specific events that may occur during the lifespan of an asset/component. Each event drives a data sustainment action within BUILDER as detailed in the figure. The remainder of this section will focus on specific actions to take when adding or modifying BUILDER's inventory (and inspections) using BRED for the following conditions:

- New building construction
- Asset replacement
- New asset installation
- Building renovations
- Asset rebuilds
- Environmental impacts



Source: Annex I of IMCOM OPORD

Figure 6: BUILDER Sustainment Decision Tree

2.2 Adding New Building Construction or a Change in Real Property Status

2.2.1 Overview

Adding a building to an installation's BUILDER inventory first requires Form 1354 to be processed and GFEBs updated. New Real Property assets meeting the Army BUILDER Business Rules will be updated in BUILDER in the quarterly HQIIS topline. Similarly, if a previously Outgranted or Leased building is being transferred to Government ownership, it should be added to BUILDER as a new building.

The screenshot shows the BUILDER software interface. On the left is a list of buildings, with '33530 - PYROTECHNIC PRODUCTION BUILDING' selected. On the right is the 'Add Building' form. The 'Building ID' field is highlighted with a red box and contains the value '33530'. The 'Quantity' field is also highlighted with a red box and contains the value '13,808.00'. The 'Year Built' field is highlighted with a red box and contains the value '1942'. The 'Building Use' is '22622 - PYRO PRODUCTION'. The 'Const. Type' is 'Permanent'. The 'Address' section includes 'St Address: 33530 XYZ STREET', 'City: Pine Bluff', 'State: AR', and 'Zip Code: 71602-9500'. The 'Point of Contact' section includes 'Name:', 'Phone:', and 'E-Mail:' fields.

Figure 7: Add New Building - Validate Building Number

2.2.2 Action Steps

1. To add inventory to a new building, a BRED file must be exported from BUILDER. Use a tablet or other portable device for use at the building. The BRED file can be exported by building and by system as needed to ensure only the desired buildings and systems are being updated in the file.
2. There are various methods to physically approach a building to add new inventory to an unpopulated BRED file, but the best method is a methodical approach. Work around the exterior of the building first and populate the systems starting with A10 and working through E10. This may take several passes to ensure all systems have been collected in accordance with (IAW) the BUILDER Inventory and Assessment Guide. Direct condition ratings and inventory comments are added during the inventory collection process.
3. The interior should be approached room by room and always in the same direction within the room (clockwise or counterclockwise) and in the same sequence, such as floor, wall, and ceiling. A consistent approach helps avoid components being missed. If more than one person is conducting the updates, it is a best practice to assign either specific areas/floors or specific systems to each person to avoid duplication of work and data.
4. Open the exported BRED file and compare the building number to the number on the building. Take a good representative photo of the building that will be added to "Images" (at the top of the screen), ensuring that the photo is attached at the building level. Refer to Appendix C.3.3 Adding or replacing photos in BRED.
5. Measure the exterior of the building to verify the SF in BRED and confirm the year built is correct. Real Property data at the Building Level in BUILDER is populated from Headquarters Installation Information System (HQIIS).

Note: DO NOT change any building-level data. Changes to building information require building-level comments and modifications to HQIIS and GFEBS and should be coordinated with the Real Property Accountability Officer (RPAO).

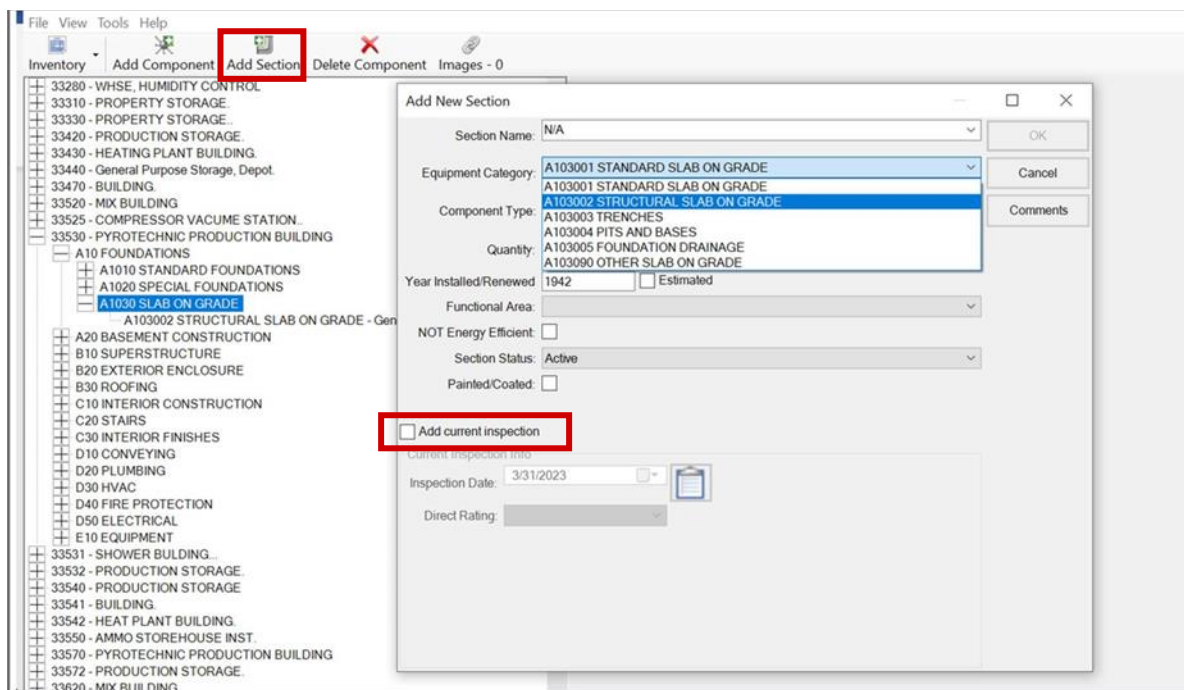


Figure 8: Add New Building - Determine Building Systems

6. Select the “+” symbol to the left of the Building number to reveal the available building systems. Starting with A10, determine if the building has a slab on grade. Click on the + next to A10 Foundations, then select “Add Section” at the top of the page. Choose the proper Equipment Category and Component Type from the dropdown menus, then add a quantity and verify the Year Installed is correct (modify the Year Installed as needed).
7. Before selecting “OK”, add a current inspection by clicking the box toward the bottom of the “Add New Section” window. The clipboard to the right of the inspection date allows the user to add the distress to the inspection comments if the rating is Amber+ or lower.

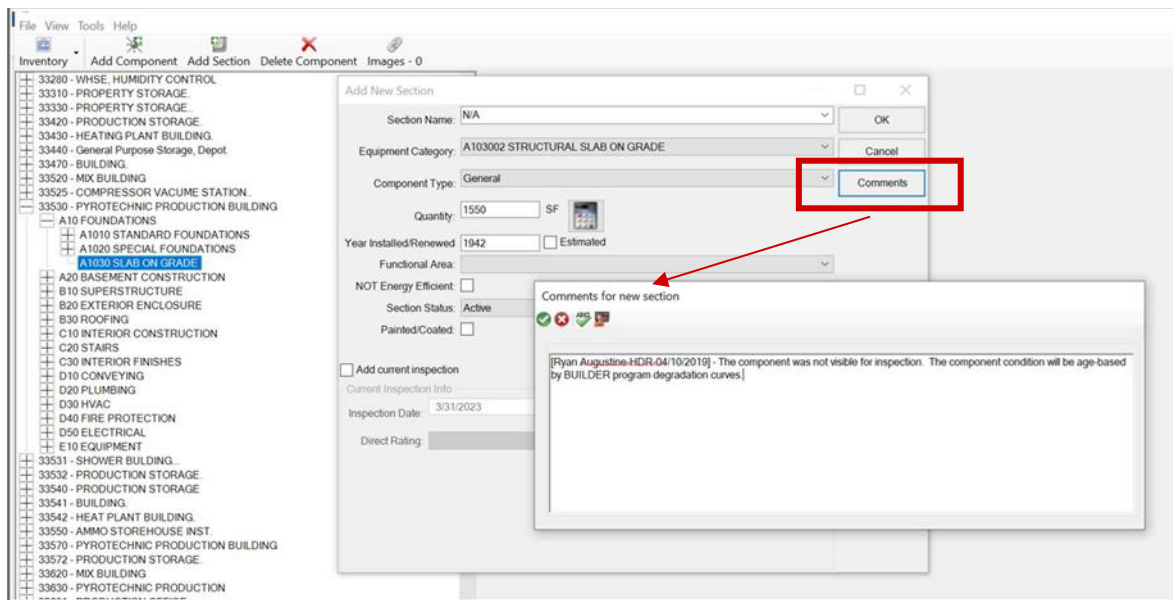


Figure 9: Add New Building - Add Comments

8. If the component is not visible for inspection or there are other section comments that need to be added, click the “Comments” box next to the “Component Type” field. (To add or edit comments after the component section has been added, click on “comments” at the top of the screen.)

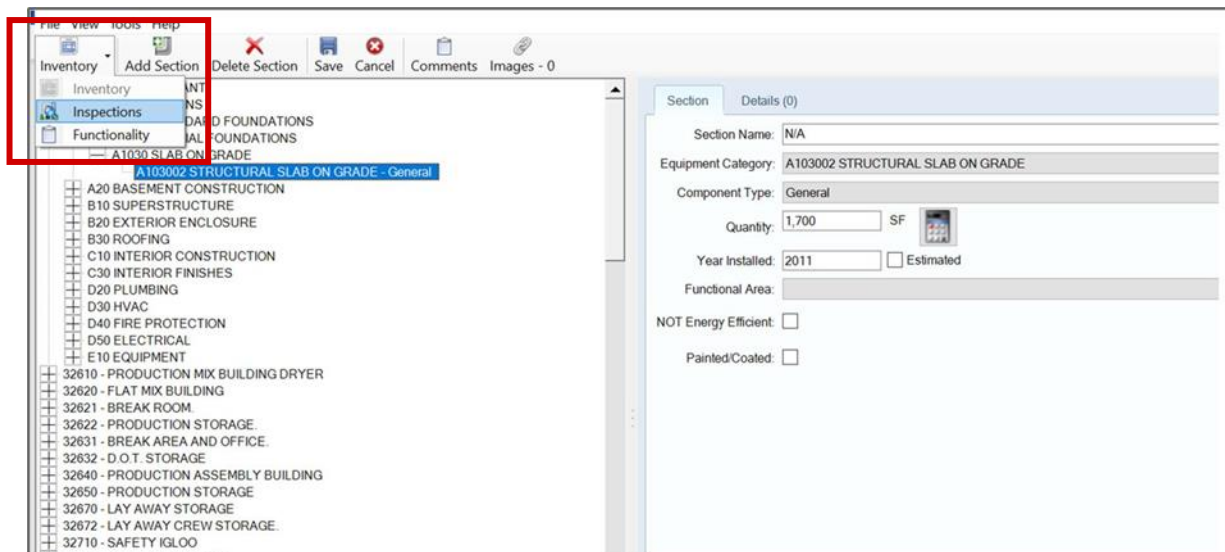


Figure 10: Add New Building - Access Inspection Page

9. If the Direct Condition Rating or comments were not added while adding the component-section information, the inspection page can be accessed by clicking on “Inventory” in the top left corner of the screen and choosing “Inspections” from the dropdown menu.
10. Per the BUILDER Guide standards, all inspections with an Amber+ or lower rating must include a photo. This is accomplished on the “Inspections” screen where the Direct Condition Rating (DCR) and/or comment can be reviewed as well. (Refer to Appendix C.3.3 Adding or replacing photos in BRED).

Proceed as described in Action Steps 1-10 above for every component in the building, following the rules provided in the BUILDER Guide for sectioning and ratings. Note that new construction (or any building not previously captured in BUILDER) should include all 14 building systems and their associated “in scope” components (per Army standards).

Note: Much of the new construction inventory can be added to BUILDER (using BRED) from the building’s as-built drawings, equipment schedules, or even Building Information Models (BIM).

2.3 Replacing Assets

2.3.1 Overview

Assets will be replaced at various times over the life of the building, and some may be replaced multiple times. It is important to keep BUILDER up to date to allow the system to generate complete and accurate metrics and work items.

It is best to have the invoice or work order on hand when entering the information on the new component, especially if the new component requires section details. Often times components are installed in such a manner that the data plate is difficult to see to record BUILDER’s required details.

In this example, an existing make-up air unit is being replaced.

2.3.2 Action Steps

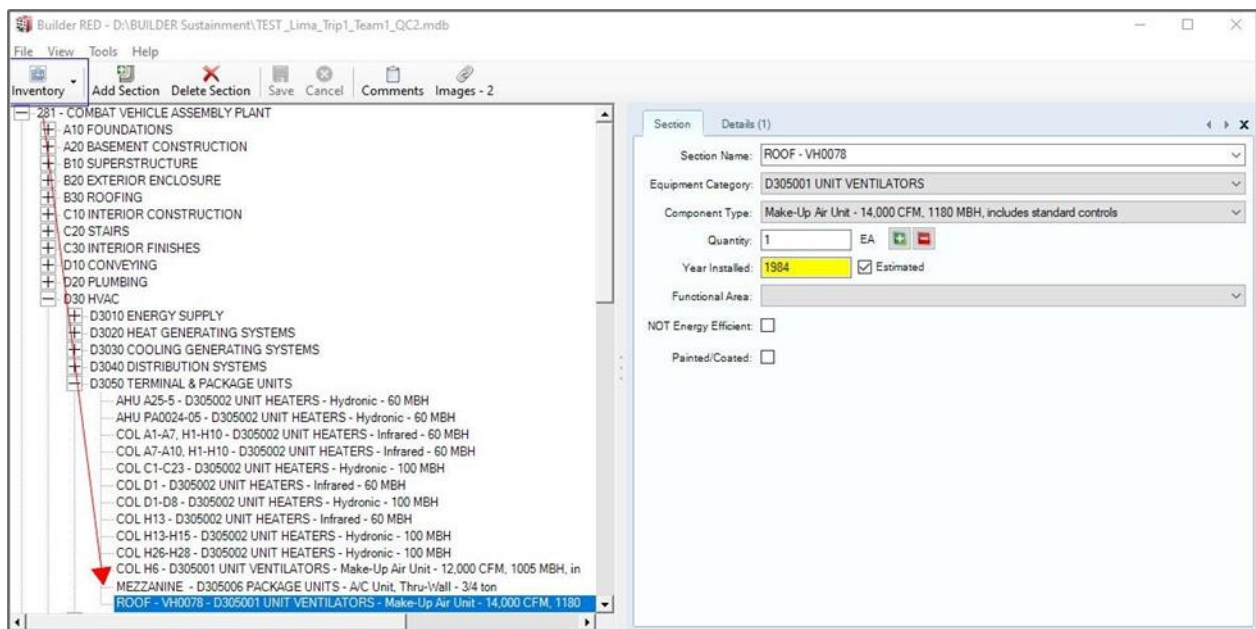


Figure 11: Replacing Assets - Find the Unit in Dropdown Menu

1. First, find the unit in the dropdown menu in BRED. Review photos and section details to verify that the correct unit is selected.

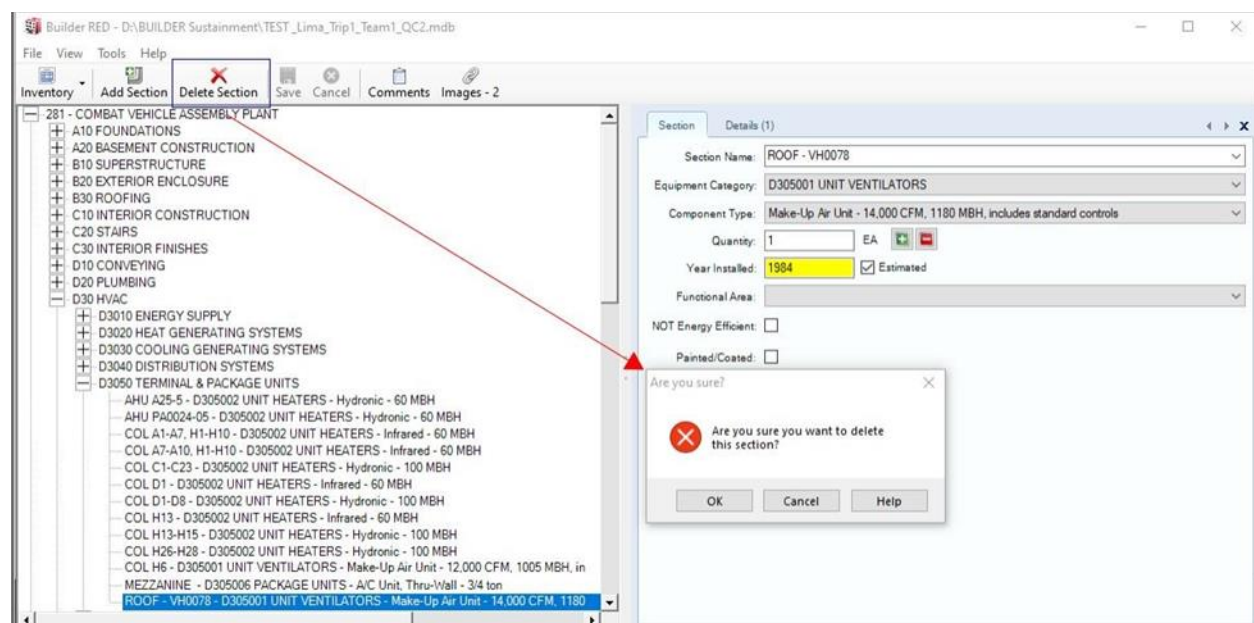


Figure 12: Replacing Assets - Delete Section

- Once the unit has been verified, click on “Delete Section” at the top of the window. A pop-up box will appear “Are you sure you want to delete this section?” Click “OK”, and the component section will drop off the list.

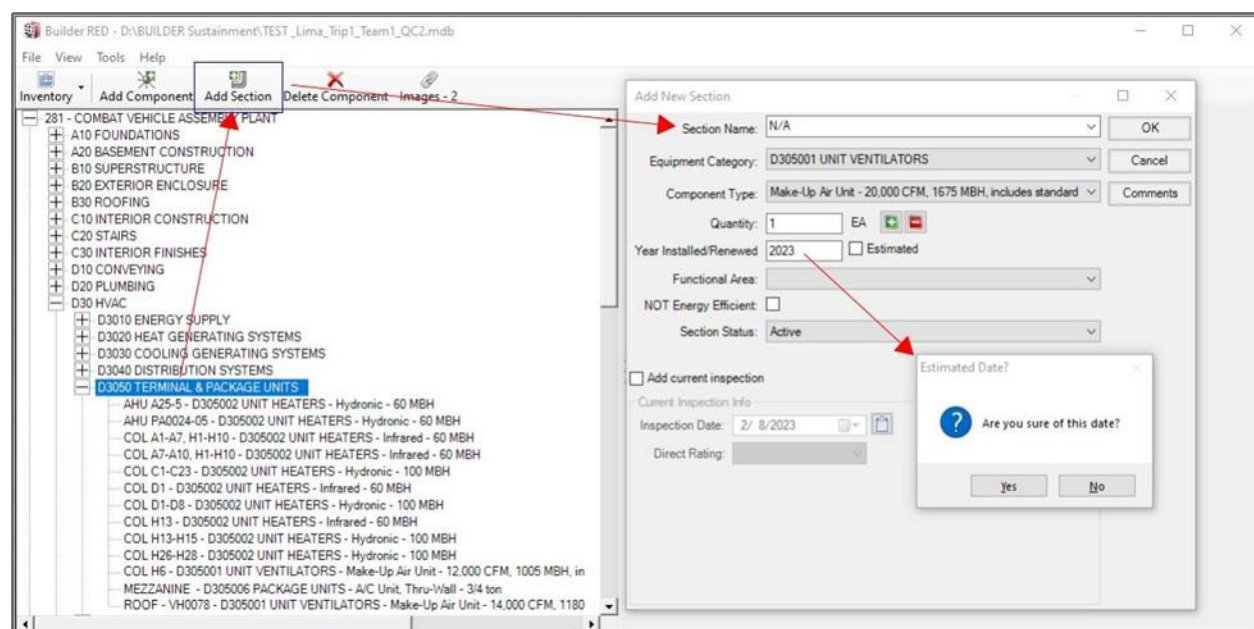


Figure 13: Replacing Assets - Add the New Component

- To add the new component, click on the system component, then “Add Section”. Choose the correct Equipment Category and Component Type from the dropdown menus in BRED, then enter the quantity and the Year Installed. Click “yes” on the pop-up box “Are you sure of this date?”
- Follow the Army BUILDER SMS Inventory and Assessment Guide to determine if section details, photos, or a Direct Condition Rating (DCR) is needed. For specific how-to instructions, refer to Appendix C.3.2 Editing an Inventory Record in BRED.

5. All new components that require a DCR will be given a G+ rating per Army standards.

Note: This is a two-step process; "Delete Section", then "Add Section." It is important to completely delete the old component before adding the new component. DO NOT simply edit the old data in BRED to reflect the new inventory since doing so would not remove the previous inspections.

2.4 Installing a New Asset

2.4.1 Overview

Changing missions or requirements in a space may require that additional components be installed. From new panelboards, air conditioning units, a new office area, or an entire addition, all new components must be added to BUILDER to keep the inventory up to date for sustainment.

Invoices, work orders, material lists, and contractor bids are all good documentation to have on hand when entering new data. Some preliminary data entry may be accomplished in the office prior to field verification, especially if the list of new components to be entered is extensive.

2.4.2 Action Steps

The screenshot shows the 'Add New Section' dialog box. It has a title bar with a close button. The main area contains several input fields and checkboxes. On the right side, there are three buttons: 'OK', 'Cancel', and 'Comments'. The 'Year Installed/Renewed' field is highlighted in yellow. The 'Current Inspection Info' section is collapsed.

Figure 14: Install a New Asset - Add Year Installed

1. Refer to Section 2.2 New Building Construction / Change in Real Property Status for the steps needed to add new sections. Keep in mind that some new components may have complementary systems that have been added to support the new component such as ductwork, piping, or branch wiring.

2. Follow the Army BUILDER SMS Inventory and Assessment Guide on Section Naming conventions for new assets when the same asset type already exists in the building.
3. When entering new data, BRED makes calculations on certain equipment types to arrive at the “Year Installed”. If the Year Installed is estimated by BRED, the box will be highlighted yellow.
4. A common error is failing to change the Year Installed and uncheck the “Estimated” box.

2.5 Entering Building Additions

2.5.1 Overview

Entering building additions into the inventory on an existing building follows the same steps outlined in Section 2.2: New Building Construction / Change in Real Property Status. However, section names will start with ADDITION on every component section in the new space and the Year Installed must be the same for all sections named ADDITION.

2.5.2 Action Steps

These action steps are described in Section 2.2 New Building Construction / Change in Real Property Status.

2.6 Entering Building Renovations

2.6.1 Overview

A renovation can be a combination of asset replacements and new asset installations. It can encompass a single room, multiple rooms, an entire floor, or an entire building. It is important to keep in mind that a renovation is not for a replacement of a few components in a space, but rather a rejuvenation of the entire space.

Invoices, work orders, material lists, and updated drawings are all good documentation to have on hand when entering renovation data. Some preliminary data entry may be accomplished in the office prior to field verification, especially if the list of new components to be entered is extensive.

All component sections for a renovation will have the same date and the same section name, whether it is a single location (e.g., BREAK ROOM) or multiple locations (e.g., OFFICES).

2.6.2 Action Steps

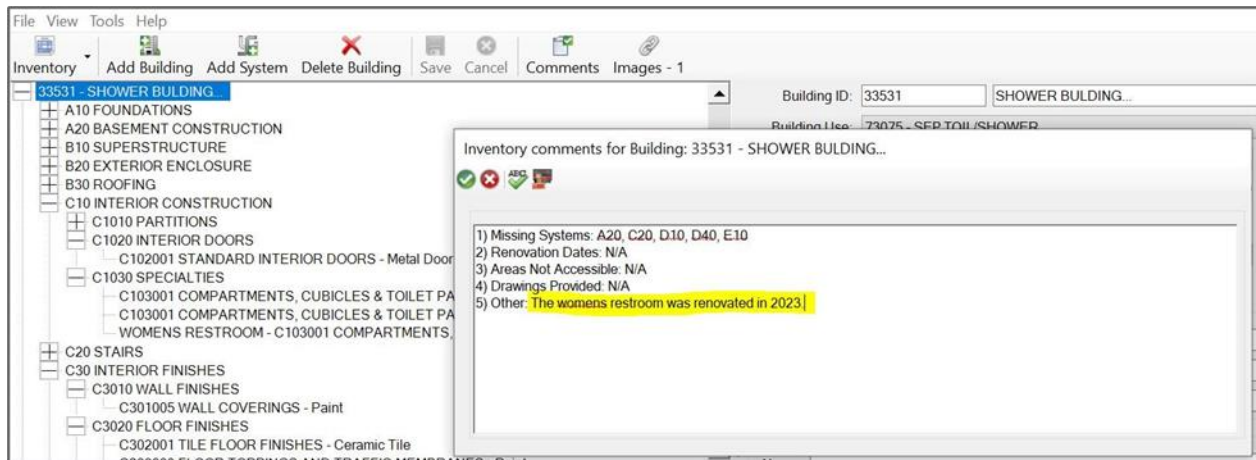


Figure 15: Building Renovations - Add Comment

This is an example of a women's restroom renovation completed in 2023.

1. First, add a building-level comment (select the "Comments" icon from the toolbar) indicating the renovation area and year while at the building level within the tree.

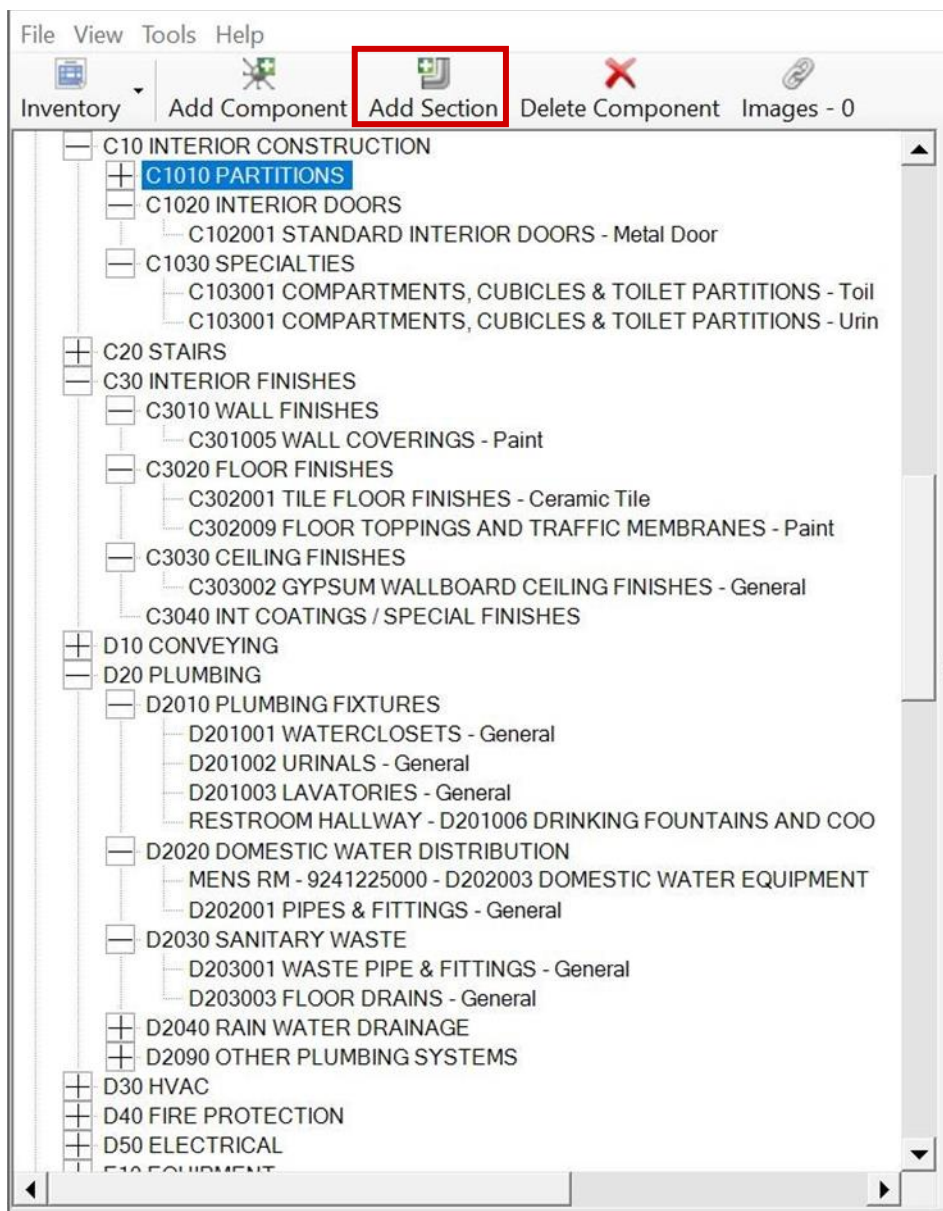
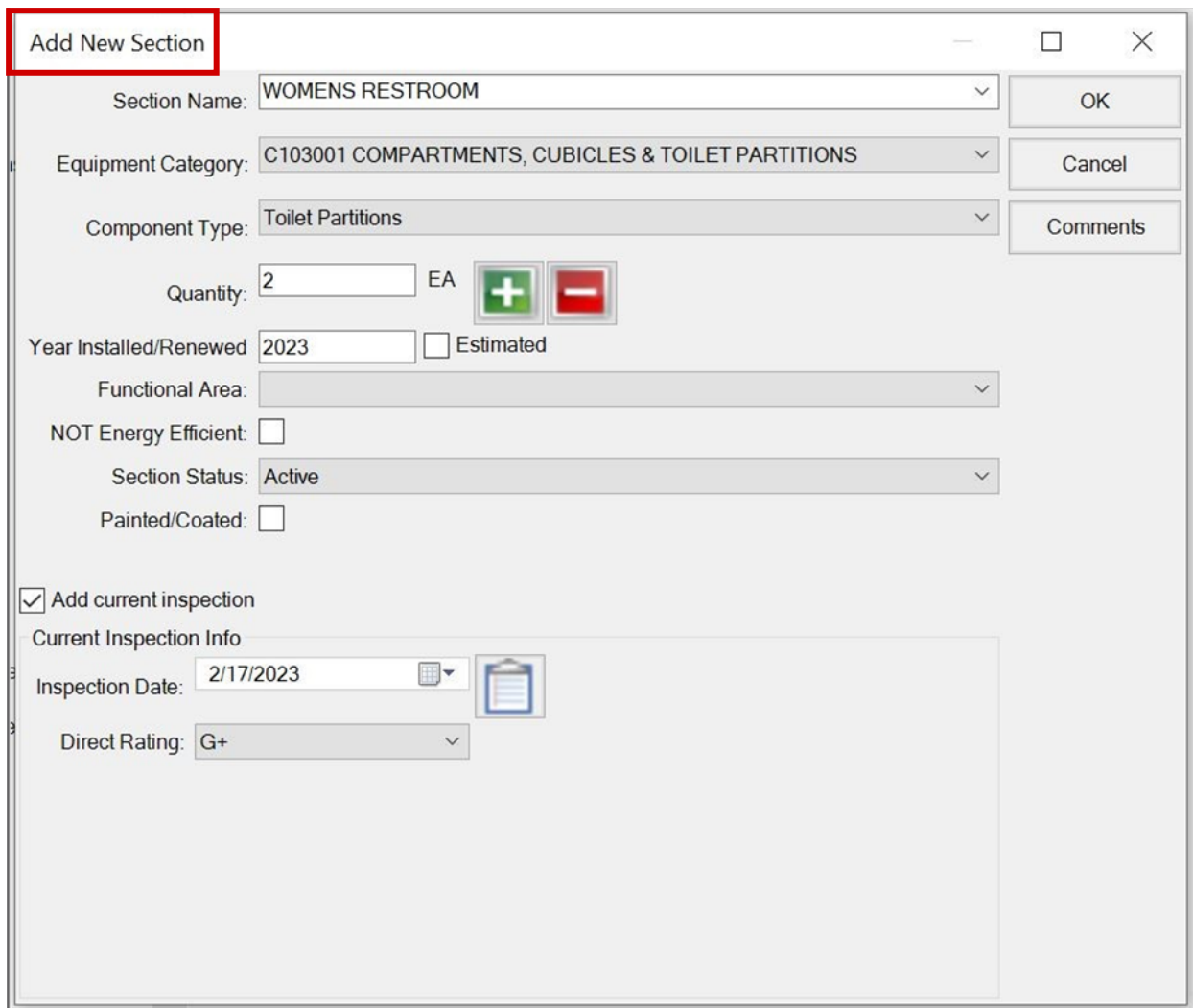


Figure 16: Building Renovations – Add Section

2. As shown in the expanded BRED tree, none of the restroom components were sectioned separately as WOMENS RESTROOM, therefore, existing data will need to be adjusted for the renovation. It is important to have a systematic approach to add new data while adjusting existing data.



Add New Section

Section Name: WOMENS RESTROOM

Equipment Category: C103001 COMPARTMENTS, CUBICLES & TOILET PARTITIONS

Component Type: Toilet Partitions

Quantity: 2 EA  

Year Installed/Renewed: 2023 ☐ Estimated

Functional Area:

NOT Energy Efficient: ☐

Section Status: Active

Painted/Coated: ☐

☒ Add current inspection

Current Inspection Info

Inspection Date: 2/17/2023  

Direct Rating: G+

OK Cancel Comments

Figure 17: Add New Section

3. Add the first new section and rating as needed, moving down through each component to cover the full extent of the renovation. Make sure to include the appropriate section name for each

The figure consists of two screenshots of a software interface, likely for facility management or inventory tracking. Both screenshots show the 'Details (0)' tab for a component named 'Toilet Partitions' under the category 'C103001 COMPARTMENTS, CUBICLES & TOILET PARTITIONS'. The 'Section Name' is 'N/A'. The 'Quantity' field is highlighted with a red box in both screenshots. In the top screenshot, the quantity is 6. In the bottom screenshot, the quantity is 4. A red arrow points from the quantity field in the top screenshot to the quantity field in the bottom screenshot, indicating a change in the inventory quantity. Other fields include 'Year Installed' (1980), 'Functional Area' (empty), 'NOT Energy Efficient' (checkbox), and 'Painted/Coated' (checkbox). The 'Estimated' checkbox is checked in both screenshots.

Figure 18: Adjust Inventory Quantity

4. Next, adjust the existing inventory quantity to update the remaining partitions that were not renovated.
5. Continue in this manner until all components for the renovation have been entered and the existing quantities have been adjusted.

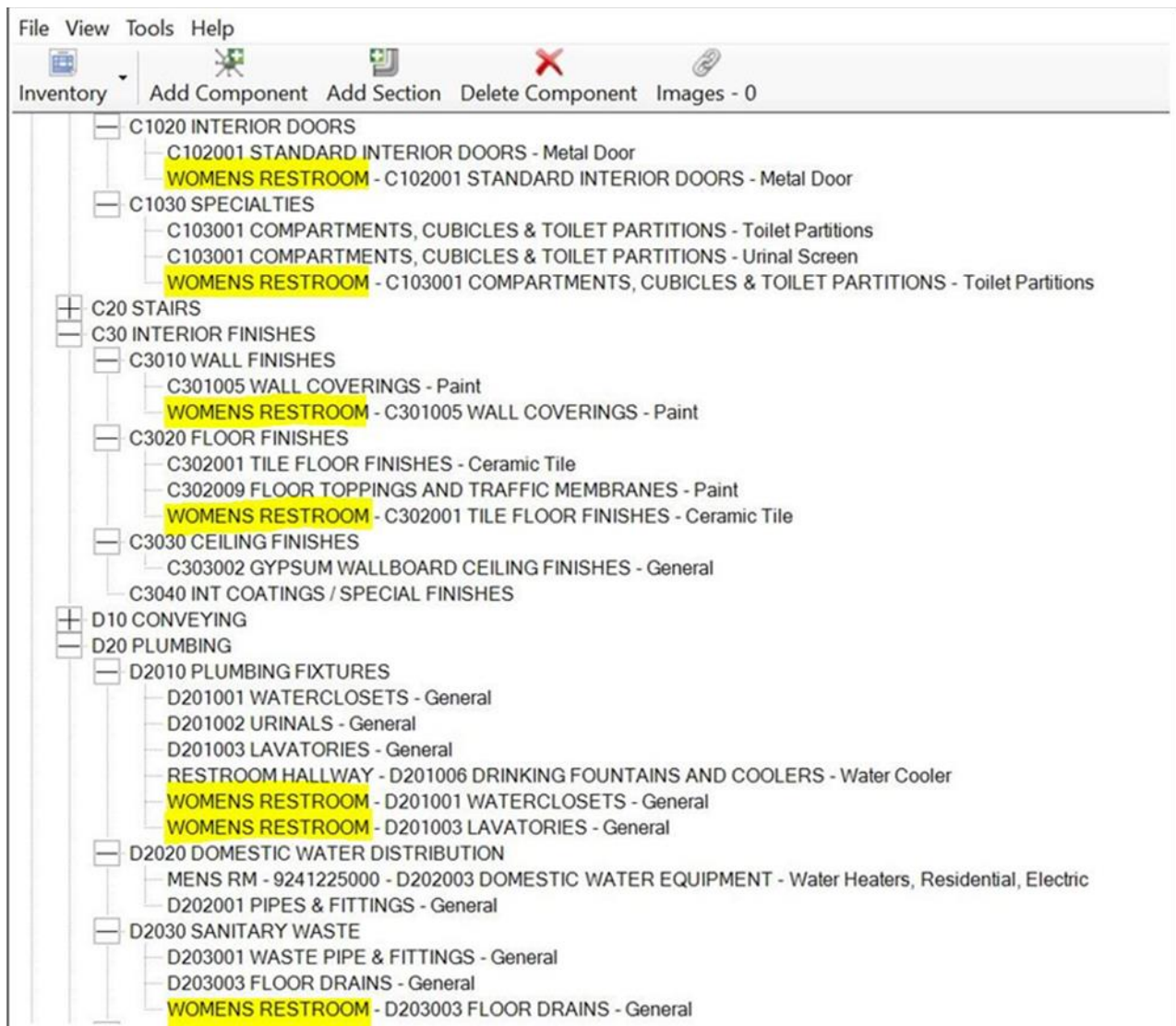


Figure 19: New BRED Inventory Tree

This is the new BRED inventory tree after the example renovation has been entered. Notice that not all equipment categories were affected. In this example, the ceiling and piping were not updated; therefore, the original data remains the same, and new sections were not added.

2.7 Entering Asset Rebuilds (inventory and inspection updates)

2.7.1 Overview

Replacing an asset is not always the best option for renewal. Replacements can be costly, and space or room configurations can also have significant impacts on the ability to replace an entire component. Rebuilding an asset is a viable option in lieu of replacement and will change the component's condition.

Pumps, boilers, chillers, and air handlers are examples of some of the mechanical equipment that can be rebuilt or overhauled.

When a rebuild has occurred, the information must be updated in BRED.

2.7.2 Action Steps

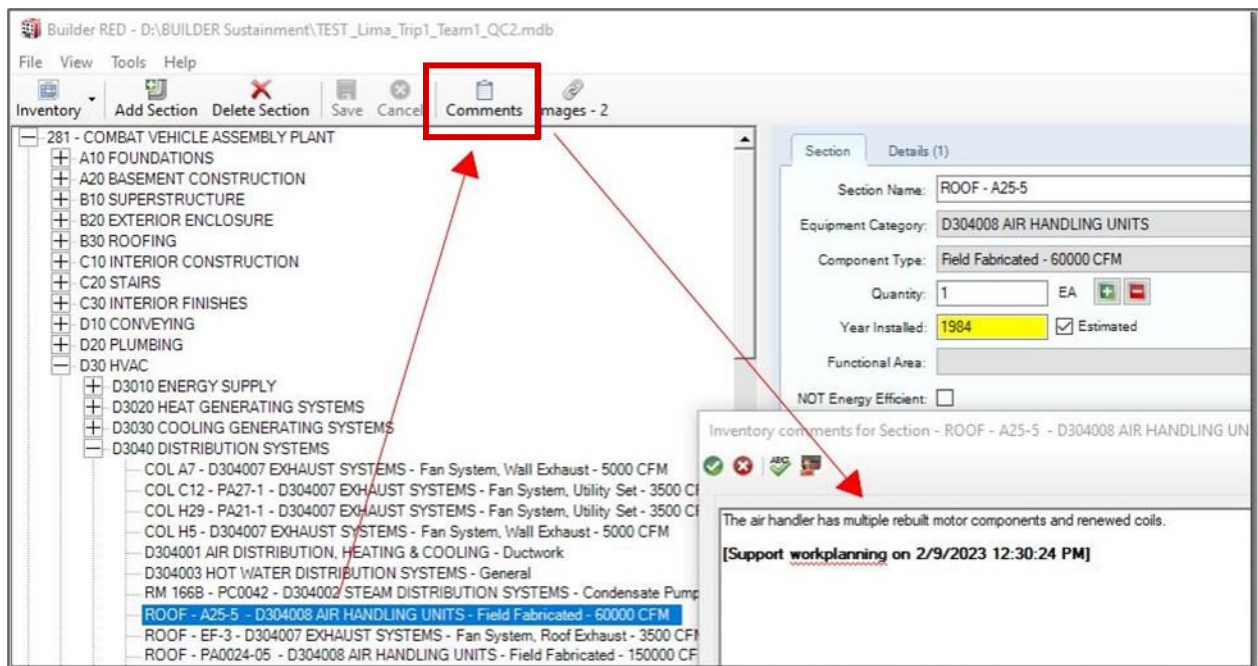


Figure 20: Entering Asset Rebuilds – Add Inventory Comments

1. Find the desired component section in the file and add inventory comments detailing the work that was completed.

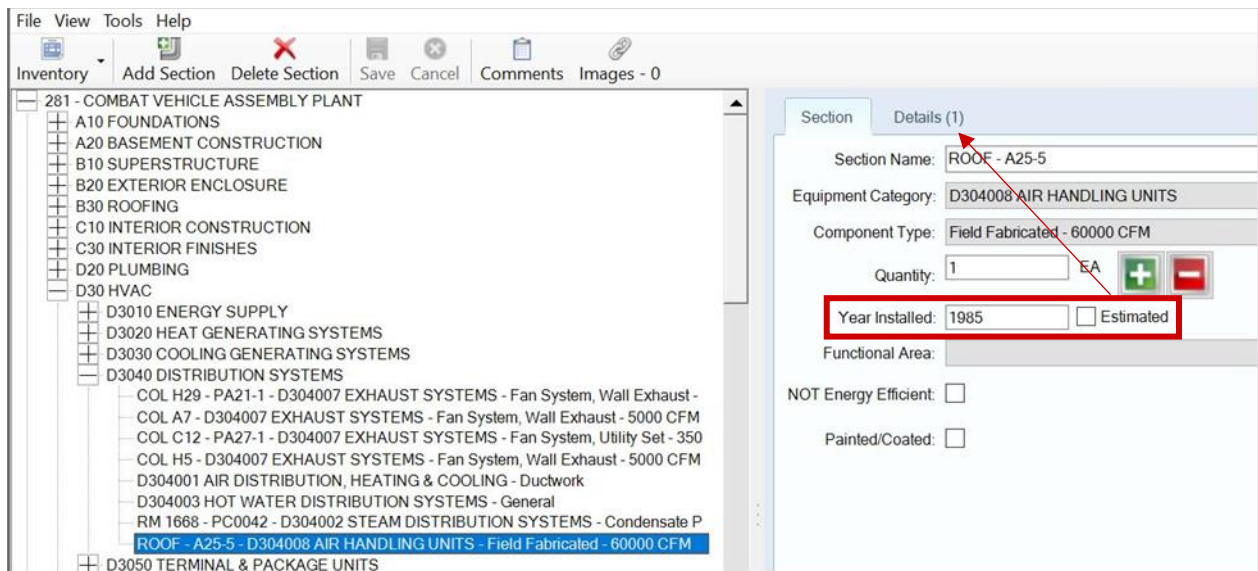


Figure 21: Year Installed

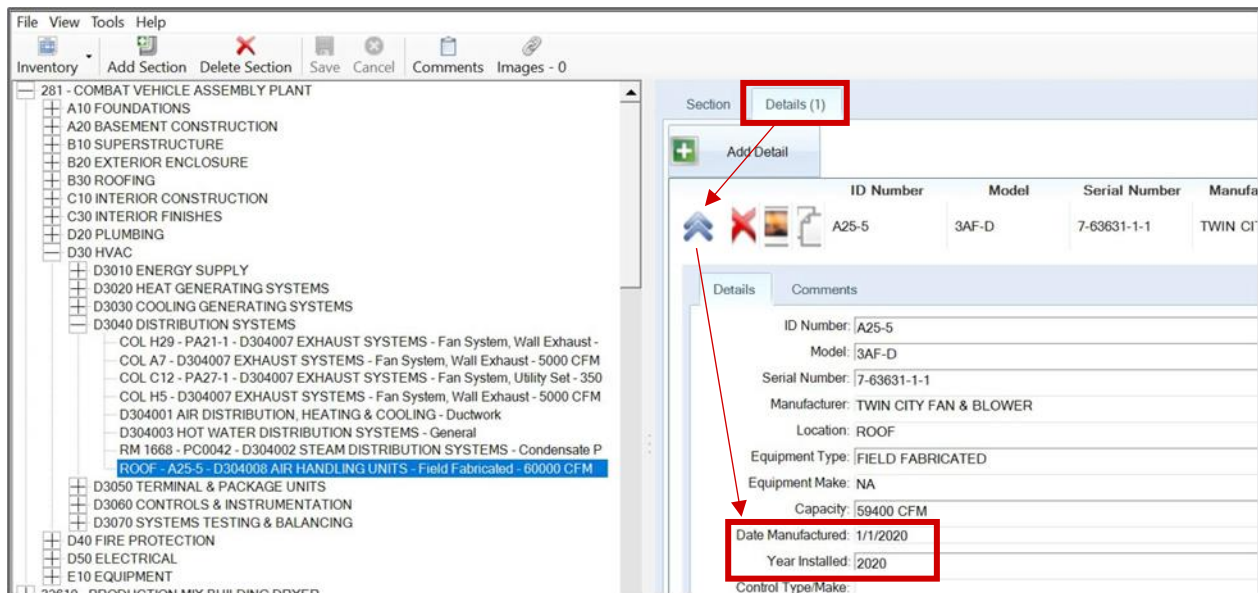


Figure 22: Details Tab

2. Update the Year Installed to the date that the work was completed and then change the year in the section details to match.

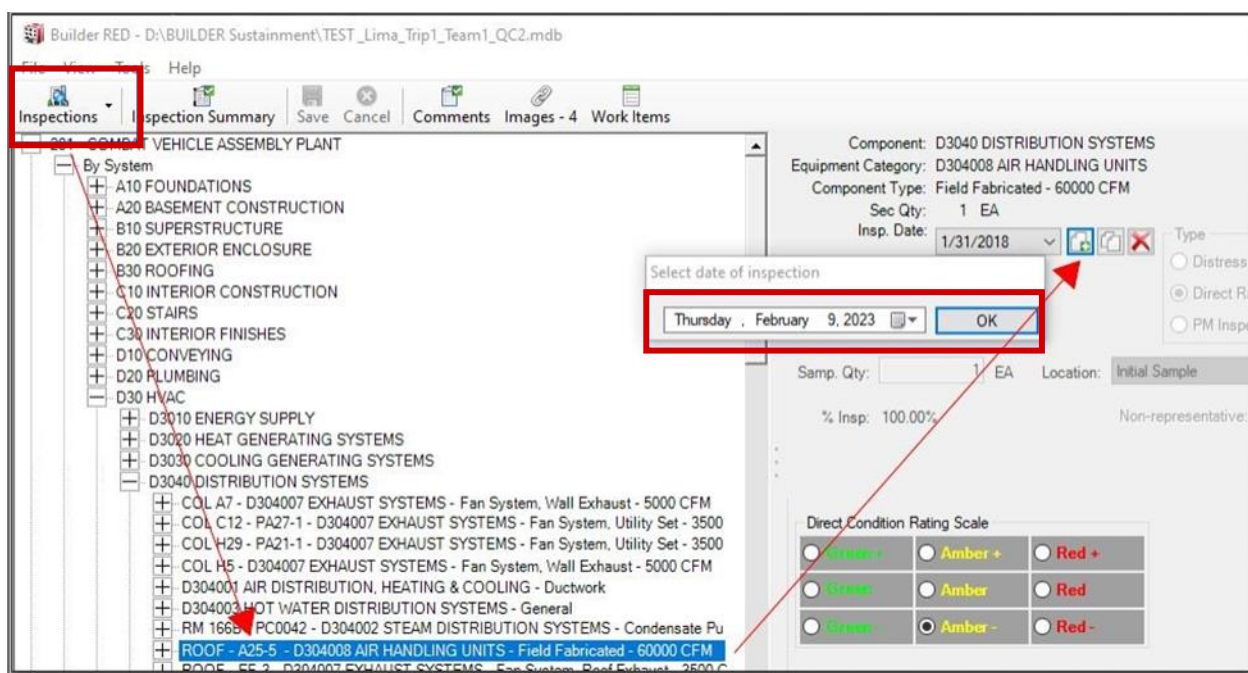


Figure 23: Inspections Tab

3. Switch to Inspections and select the date of the rebuild and add a new G or G- inspection, aligned with the DCR guidance in the BUILDER Inventory and Assessment Guide and the extent of the rebuild.

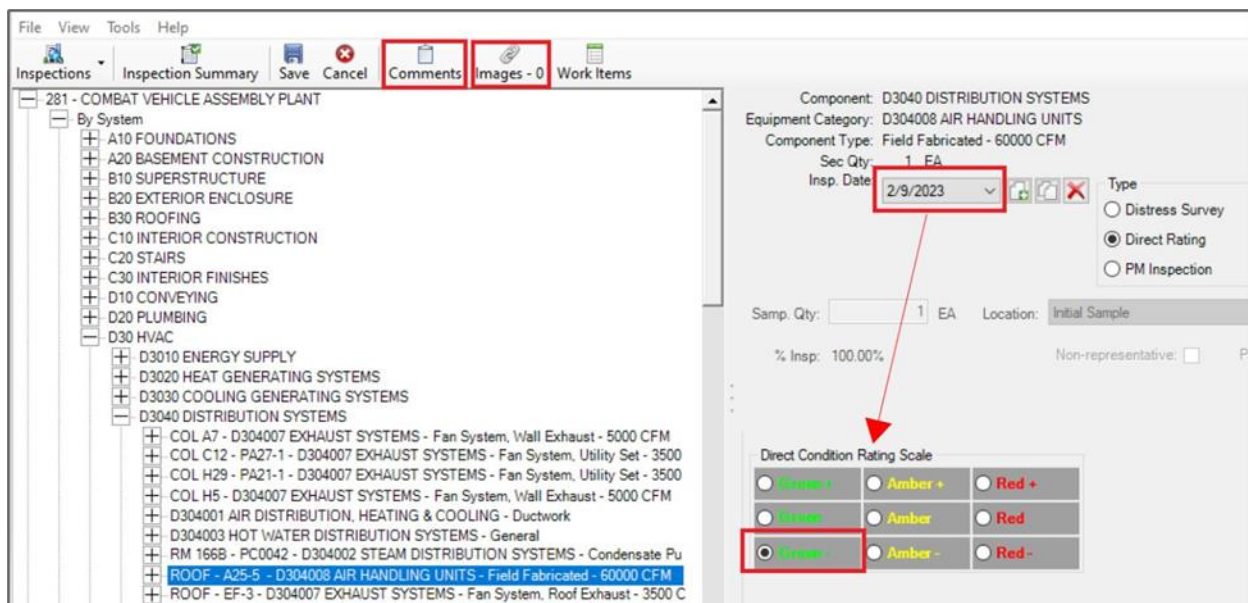


Figure 24: Comments and Images

4. Then add Comments and Images to the new inspection event.

Many repairs can also fall into this category as a renewal that can change component condition. Missing roof shingles, broken acoustical ceiling tiles, broken light covers, missing blanks in panelboards, and damaged walls are just a few examples of distresses that can be repaired to change the component rating for the better. Read the inspection comment carefully to verify that the documented distress was

remedied by the repair before adding a new rating, and then follow the steps as outlined above if a significant enough portion of the component was repaired to qualify as an Asset Rebuild.

2.8 Entering Environmental Impacts (inspection updates)

2.8.1 Overview

Seasonal sun, snow, ice, and rain as well as extreme weather events such as hurricanes, tornadoes, flooding, hail, and ice storms can contribute to the premature deterioration of building components. Depending on the severity of the season or event, roofs, doors, and other exterior components should be monitored after each event or extreme season.

Interior components can be affected by the environment as well. Look for improperly sealed exterior windows, poorly functioning HVAC units and dripping pipes which can all cause moisture buildup or other degradation on the inside of the building.

It is important to monitor conditions after any event and update the component condition as needed. When changes to condition do occur, add a new rating to BRED.

2.8.2 Action Steps - Reviewing a DCR in BRED

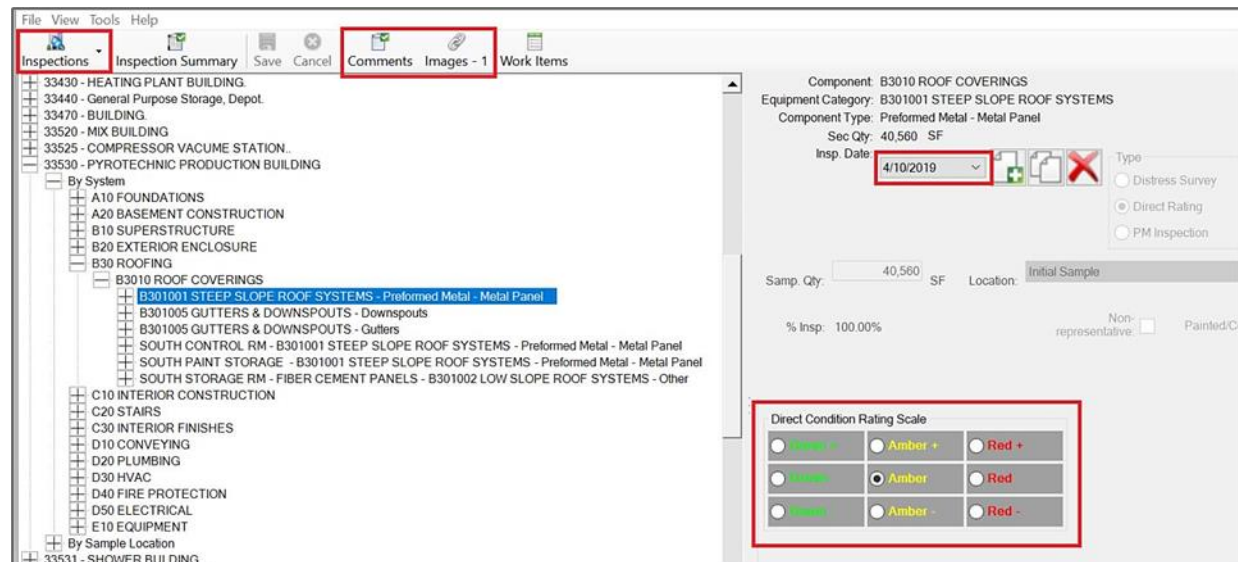


Figure 25: Inspections Tab

1. To determine if there has been a change in condition, click on Inspections to view the current Direct Condition Rating and the date it was entered. Comments and Images (if any) must also be reviewed to determine if a new inspection is needed.
2. If a change in condition has occurred, add a new inspection with comments and a new photo (if the rating is Amber+ or lower).

2.8.3 Action Steps - Adding a New Inspection

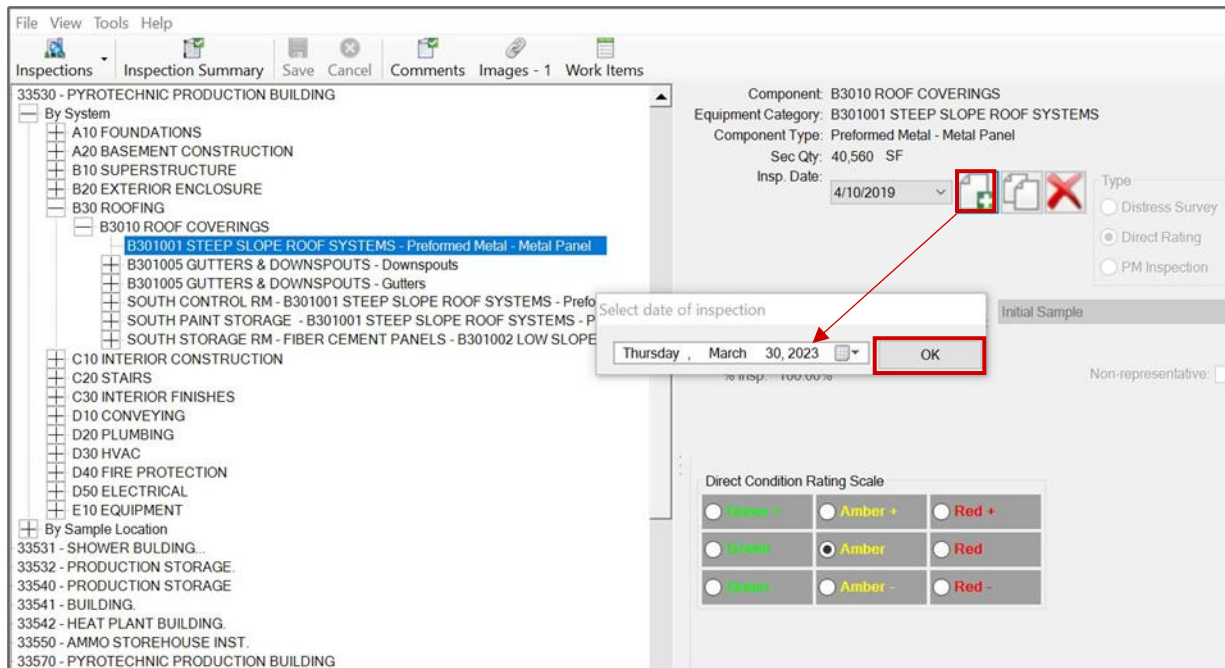


Figure 26: Adding a New Inspection After an Environmental Impact

This example shows how to add a new Red inspection after an environmental impact.

1. Click on the “+” icon.
2. The current date will appear. Click OK.

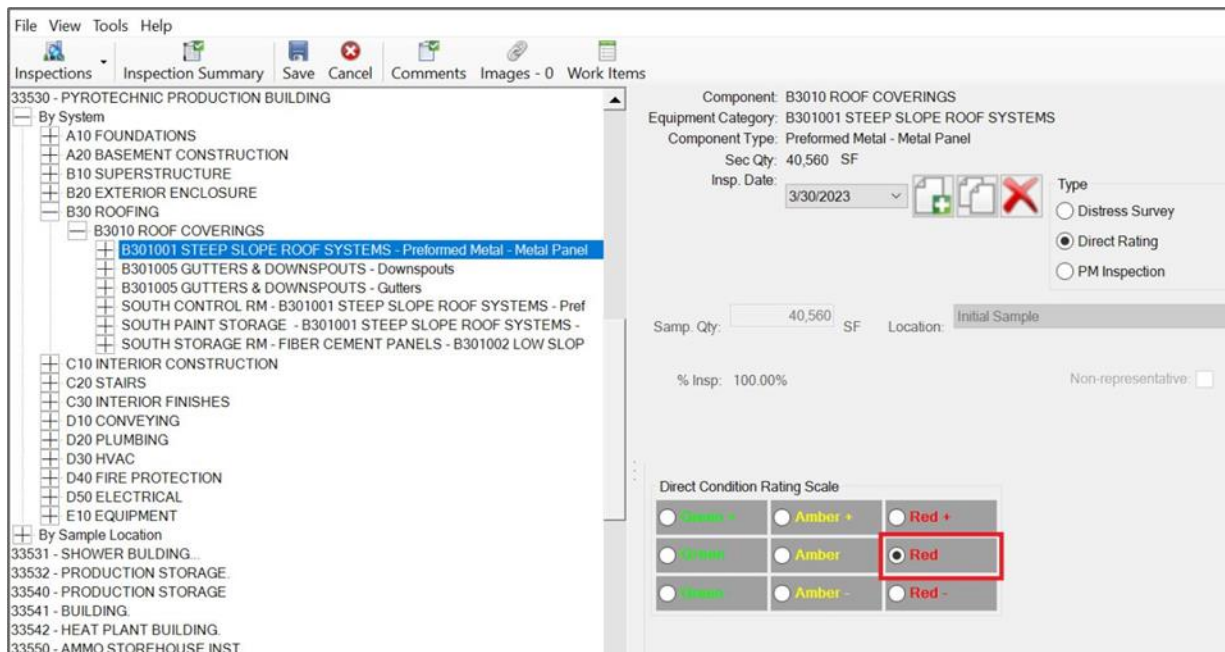


Figure 27: Red Inspection Radio Button

3. Click the radio button next to Red.

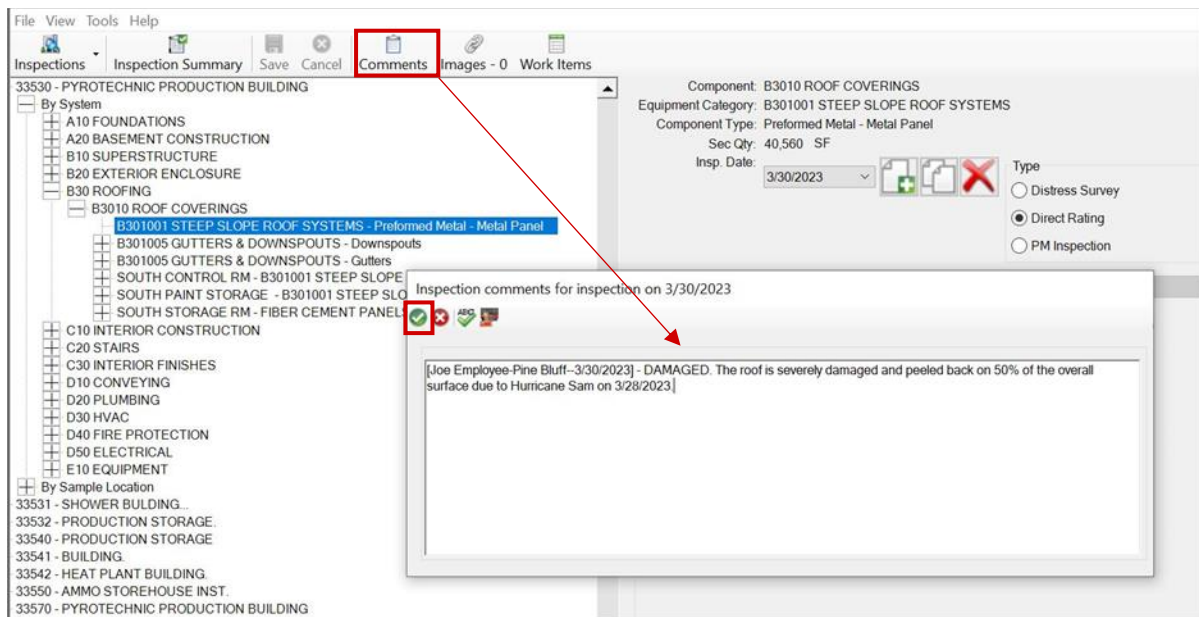


Figure 28: Add a Comment

4. Add a Comment, following the BUILDER manual guidelines for proper distresses and comment structure.
5. When the Comment is complete, click the green check mark.

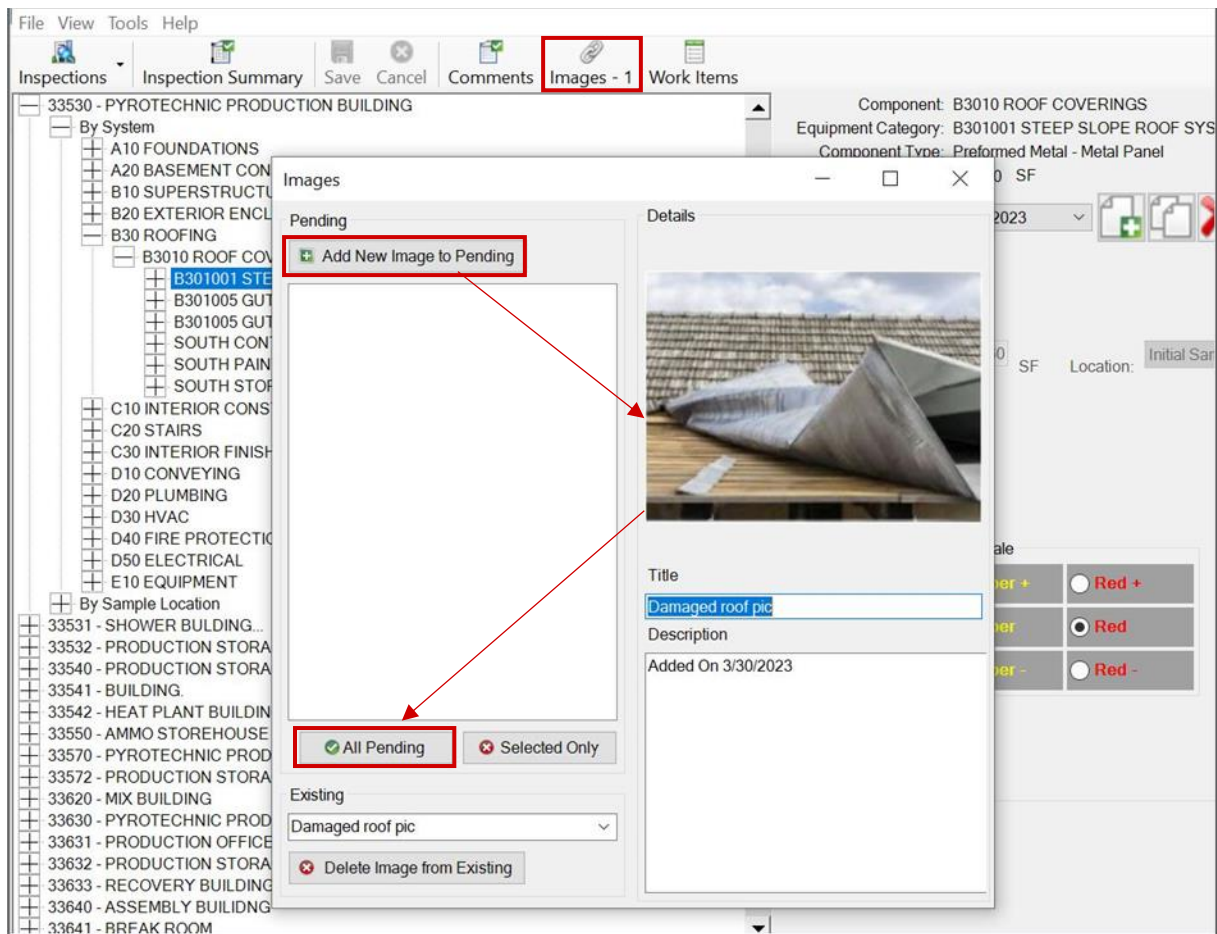


Figure 29: Add Photo of Distress

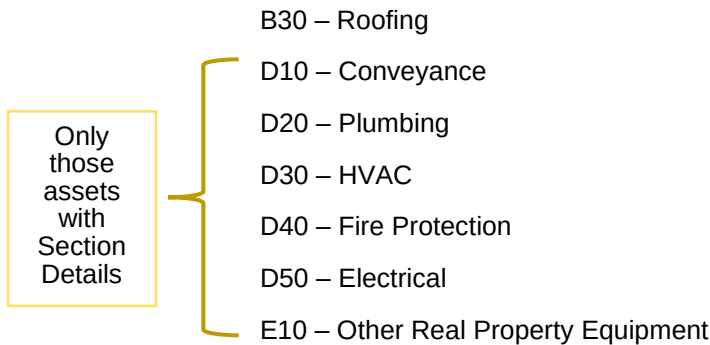
6. Add a photo showing the distress by clicking on Images. Choose the photo from your gallery, and then click All Pending.

Updating BUILDER data after normal preventative maintenance or service calls is another critical aspect of sustaining the BUILDER data. BUILDER is designed to provide building maintenance technicians with the ability to update building data as they are performing normal preventive maintenance rounds or responding to service requests. BRED is a standalone field tool designed for use with BUILDER that does not require an internet connection while working in the field.

Reference Section 3.4 of the DCS, G-9 Sustainment Guidance for additional suggestions of when BRED should be used for BUILDER sustainment.

3. Prioritizing BUILDER Reassessments

With the initial BUILDER implementation complete, JMC Installations shall implement a prioritized approach to reassessment of 100% of their critical facility inventory over five years to minimize cost and resource impact on manpower constrained DPWs. Assessments of existing facilities shall be limited to the seven critical building systems as follows:



Note: For D10 through E10, only components that require Section Details per the Army BUILDER SMS Inventory and Assessment Guide should be included in the data sustainment plan.

3.1 Prioritizing JMC BUILDER Sustainment requirements

Keeping the BUILDER data relevant and effective JMC Installations shall consider a mission-oriented, risk-based approach to prioritizing their BUILDER sustainment activities.

To reduce the cost of reinspecting 100% of the facilities at each installation, additional criteria have been developed and adopted to focus on the most mission-critical facilities and systems in the inventory.

3.2 JMC BUILDER Sustainment Guidelines

Installation BUILDER POCs shall follow the BUILDER Sustainment Decision Tree (Figure 30) to identify a list of facilities to be reassessed with the goal of 100% of the facility inventory (that meets the Decision Tree criteria) being inspected once every five years. New inventory (new construction, renovations, etc.) and completed work items shall be updated annually and should include all 14 building systems and associated components as is detailed in Section 2 of this Guide.

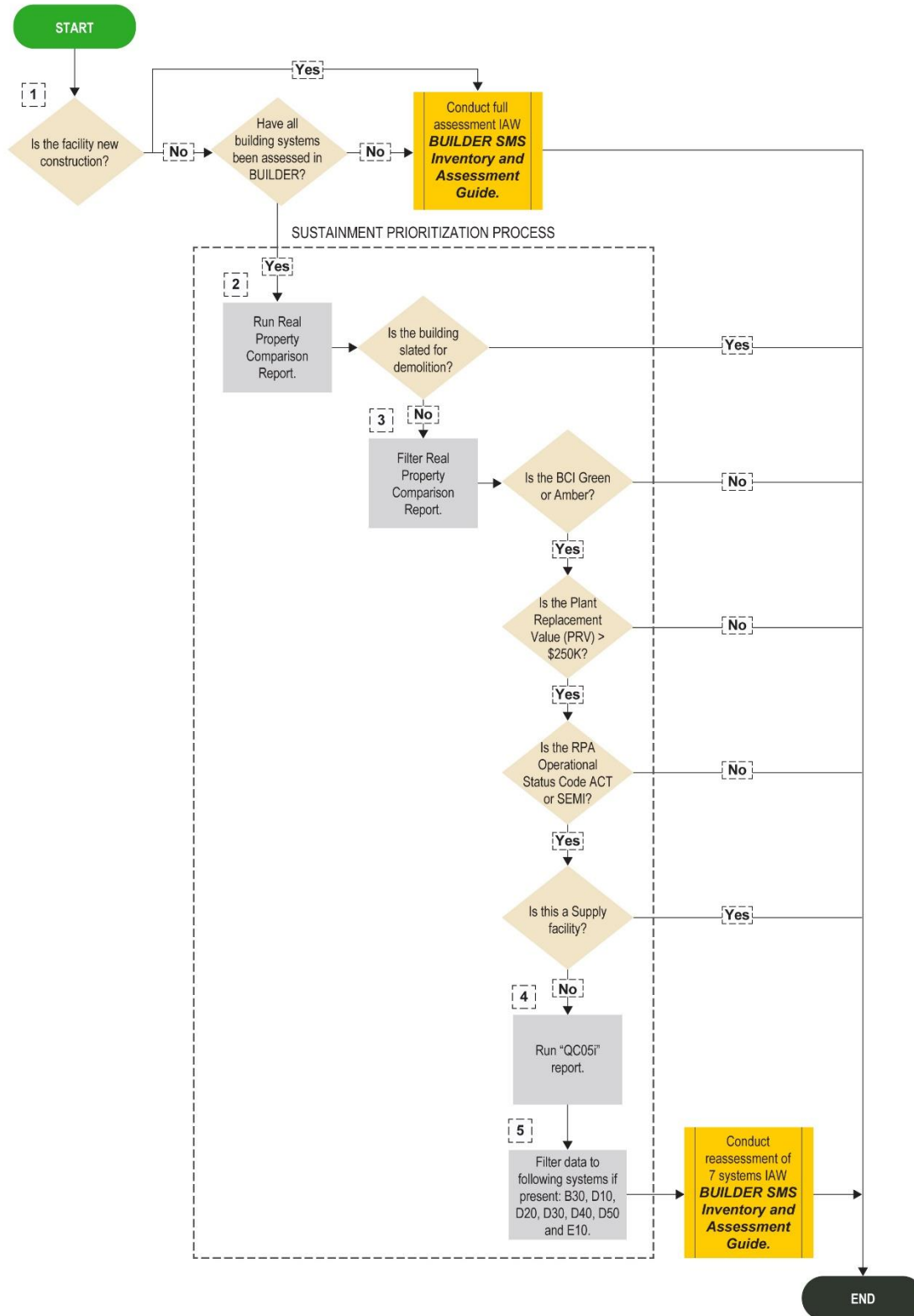


Figure 30: JMC BUILDER Sustainment Prioritization Process

3.3 Identify Facilities for Sustainment Activities (Reassessment)

3.3.1 Overview

The qualifying events outlined in Section 2.0 describe the various scenarios in which BUILDER facility inventory shall be updated. This Section focuses attention on developing a prioritized list of existing facilities requiring Sustainment Activities (reassessment) over a 5-year period.

3.3.2 Action Steps

1. "Sustainment activities" assumes the building has been previously assessed in accordance with (IAW) the BUILDER SMS Inventory and Assessment Guide. If the building is new to the inventory, proceed with a "full" assessment (all 14 systems) of the entire building. (Adding new construction to BUILDER via BRED is outlined in Section 2.2 of this Guide). Therefore, Step 1 is to navigate to the "Custom Reports" menu in BUILDER (at the Site level) to run the Real Property Comparison (RPC) report and export it to MS Excel. (See Appendix D for details on running reports in BUILDER.)
2. Coordinate with Facilities Engineering to determine if the building is programmed for Demolition. If yes, exclude it from BUILDER sustainment activities. If not, continue with Step 3.

RP Accountable Sub-Organization Code	Building Number	Building Name	RP Category Code	Category Code	Category Code Description	RP Facility Year Built	Year Built	RP Interest Type Code	RP Operational Status Code	Building Status	RP Plant Replacement Value	Building PRV	FCI	BCI	A19
READINESS	00000	ENLISTED UNACCOMPANIED PERSONNEL HOUSING	F2111	F2111	ENLISTED UPH	2011	2011	PEE	ACT	Active	\$65,973,584	\$65,973,584	96	97	96

Figure 31: RPC Report with Category Code, RP Operational Status Code, PRV, and BCI columns highlighted in yellow

3. Using the "Filter" tool in Excel, select the following from the Real Property Comparison report:
 - 1) BCI of Green (86-100) and Amber (70-85) scores (remove Red (0-69) and Blanks)

Note: A blank BCI indicates the building requires a full assessment (or a more thorough analysis of the data) is required. Contact AMC G-4 SMS Support for assistance in these cases.
 - 2) Plant Replacement Value (PRV) greater than \$250k
 - 3) RPA Operational Status Codes Active (ACT) and Semi-active (SEMI) only
 - 4) All Facility types (Category Code Description) to include Maintenance and Production, Operations and Training, Medical, Administrative, Housing and Community Support
 - a. Exclude all "Supply" facilities with the following Category Codes as shown in Table 1: Supply Facility Category Codes to Exclude.

Table 1: Supply Facility Category Codes to Exclude

CATCD	Real Property Asset
14133	Shipping And Receiving Building
14140	Care And Preservation Shop
14150	Box And Crate Shop
14160	Blocking And Banding Facility
14172	Storage Facility, Secure
41210	Liquid Storage Nonpropellant
44110	Storage Building, General Purpose, Depot Level
44130	Controlled Humidity Warehouse, Depot Level
44135	Hazardous Material Storage, Depot Level
44150	Flammable Material Storehouse, Depot Level
44160	Radioactive Storage Warehouse, Depot Level
44180	Open Warehouse, Depot Level
44181	Vehicle Storage Facility, Depot Level
44182	Vehicle Storage Building, Depot Level
44217	Acetylene Storage Facility, Installation
44220	Storage Building, General Purpose, Installation
44222	Storage Shed, General Purpose, Installation
44223	Arms Building For Battalion And Above
44224	Organizational Storage Building
44226	Covered Storage Building, Installation
44228	Hazardous Material Storage Building, Installation
44230	Controlled Humidity Warehouse, Installation
44240	Flammable Material Storehouse, Installation
44260	Radioactive Storage Warehouse, Installation
44262	Vehicle Storage Shed, Installation
44263	Vehicle Storage Building, Installation
44288	Installation Storage Other Than Depot Or Organizational
45110	Open Storage Area, Depot Level
45210	Open Storage Area, Installation
74080	Self-Storage Rental Facility
14121	Missile Launcher And Storage Building
21640	Inert Storage (Munitions Operations Support) Facility
42107	Earth Covered Magazine (ECM)
42181	Above Ground Magazine
42184	Service Magazine
42186	Ammunition Storage Building, Depot Level
42235	Ready Magazine, Installation
42283	Ready Munition Storage
42311	Liquid Propellant Storage, Ammunition, Tank
42510	Ammunition Storage Pad

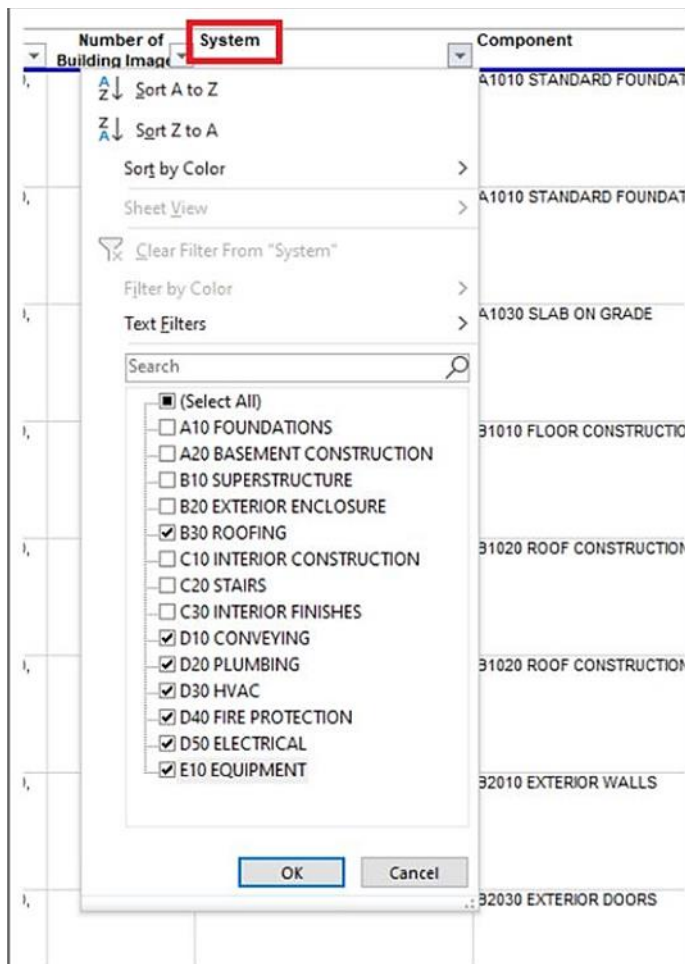
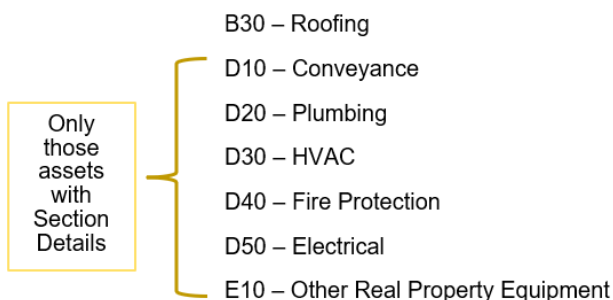


Figure 32: Screenshot of the Filter in the QC05i Systems

4. Navigate back to the Custom Reports menu in BUILDER, select the QC05i report, Run the report, and Export to Excel (the report must be run at the Site or Complex level within BUILDER).
 - a. As shown in Figure 32: Screenshot of the Filter in the QC05i Systems, manually filter for all buildings identified in Steps 1-3.
 - b. Filter out all systems except:



- c. Refer to the BUILDER SMS Inventory and Assessment Guide to determine which components require Section Details as shown in Figure 33 Example of a Component Requiring Section Details.

Detailed Inventory Guidance and Component Type Breakdown									
D30 HVAC - D3050 TERMINAL & PACKAGE UNITS									
D305003 FAN COIL UNITS - Duct Mount, 2 Pipe - 2 ton, Elec. Heat Typical Application and General Component Guidance: This component is used to inventory hydronic duct-mount fan coils that have electric backup heat. Business Rules/General/Lessons Learned/Reinspection Lesson Learned Do not confuse two-pipe hydronic systems with DX systems. There is a component type under 'D305003 FAN COIL UNITS' for DX fan coils.									
Component Type	In Scope?	Details Req?	Inventory Pic?	Group Cmmt?	Age OK?	Design Based?	Life	UOM	
Duct Mount, 2 Pipe - 2 ton, Elec. Heat	Yes	Yes	Yes	No	Yes	No	15	EA	

Figure 33: Example of a Component Requiring Section Details

5. Conduct reassessment IAW the BUILDER SMS Inventory and Assessment Guide of the resulting buildings and components.

All required reassessments should be completed within five years.

Per DCS, G-9 guidance, Installations must complete reassessment on at least 20% of the inventory annually.

Appendix A. BUILDER Data Quality Control Procedures

A.1 Quality Control

The goal of QC is to maintain data integrity, consistency, and confidence by ensuring that all business rules are adhered to and the data is complete and accurate. Key areas of focus include ensuring that the building inventory is complete, and all systems are accounted for, dates are accurate or reasonably estimated and quantities are within tolerance. To perform QC, you must be very familiar with the Army Inventory and Assessment Guide (June 2019) and the component catalog.

A.2 How to Conduct a QC Review

A.2.1 Step 1: Building-Level Review

A.2.1.1 Real Property Validation

Ensure all real property information contained in BUILDER is Real Property Information Model (RPIM) compliant. Also, ensure it complies with the use of GFEBS as the only approved system of record for the Army which feeds HQIIS. This can be accomplished by comparing the real property data from GFEBS to the BUILDER Final 09 Building Systems Quick Check report to validate:

- Real property Building list matches BUILDER data
- Building Status
- Building Area (SF)
- RPUID match
- Category Code
- Year Built
- Number of floors

Note: See Appendix D for guidance on running reports in BUILDER.

A.2.1.2 CRV vs PRV comparison

This review can be accomplished using the **Final 01 report** from BUILDER Custom Reports.

Component Replacement Value: Computed based on assets present in BUILDER; Potential flag if the value is an order of magnitude higher or lower when compared to PRV.

Plant Replacement Value: PRV is top loaded into BUILDER from HQIIS. For the purposes of reviewing BUILDER data, this check applies to vertical buildings.

A.2.1.3 Building Comments

Each building should have a comment at the building level that covers:

- **Missing Systems:** None or list as: A20, C20, D10, etc.
Note: Be sure to list them in order with a comma and space between each system.
- **Renovation Dates:** None or date renovations were completed.
- **Areas Not Accessible:** None or explain why areas are inaccessible (i.e., SCIF, under construction, HAZMAT present, etc.).
 - Include comment on roof accessibility.
- **Drawings Provided:** Yes/No or other.
- **Other:** None or add any additional pertinent Building-Level comment.

A.2.1.4 Building Photos

Each building should have at least one exterior photo that shows the entire front elevation.

A.2.2 Step 2: Building Systems Review

The BUILDER Final 09 report should be used to validate that all systems are accounted for in the data. Each system will receive a System Condition Index (SCI) score if data exists for the system. If no data exists, “N/A” populates in the report. This should be compared to the Building comments to make sure the “missing systems” listed in the comments match the data. For example, a single-story building will not have Stairs or Conveyance, so the Building Level comments should indicate that, and the SCIs for those systems should be “N/A”.

Other checks to be made include:

- **All buildings have basic systems.** It should be rare for a building to not have these systems unless the usage and/or building comment explain their absence. These include:
 - A10 FOUNDATIONS
 - B10 SUPERSTRUCTURE
 - B20 EXTERIOR ENCLOSURE
 - B30 ROOFING
- **Ensure complementary systems exist.** For example, if a building has D30 HVAC, then it must have D50 ELECTRICAL.
- **Review SCI score.** For buildings where known conditions exist, good or bad, the Final 09 provides a quick look at the condition of each system within a building to validate against known conditions.

A.2.3 Step 3: Building System Component Review

When reviewing building components, the QC05i report from BUILDER Custom Reports should be used. This report can be downloaded in Excel for an entire site or a single complex. The QC05i contains all buildings and system component-section information, section details as well as the most recent inspection. It provides a holistic look at each building and is the primary information source for a deep-dive QC.

The following steps should be included when reviewing the QC05i.

A.2.3.1 Component Inventory

- Are all systems accounted for? Filter to a single building and verify it has data for all expected systems. This can also be accomplished with the Final 09 report as a quick review.
- Are all system components accounted for? Within each system there will be a group of basic system components most buildings should include. Many of these are hidden and often missed by assessors, such as below-grade foundation components like strip footings and spread footings. Superstructure is also hidden in many buildings. Building utility systems piping, wiring, and ductwork are often missed.
- Are complementary systems complete? Since components are entered individually it is important to make sure the systems are complete. Some examples include:
 - If there is a chilled or heating hot water system with circulation pumps, there will also be piping.
 - If there is an Air Handler, there should be ductwork.
 - A hot water boiler system will usually have expansion tanks and air separators.
 - Most roof covering will have a deck that matches the roof SF quantity.
 - If you have grade beams, then you will have pile caps and if you have pile caps you will have piles.
- Are all component selections in scope based on the BUILDER manual? Most of the incorrect entries are items that are present in the building; however, the Army only wants certain component selections used. Common incorrect entries include selecting standard slab on grade, which is out of scope, the Army uses structural slab on grade. Another common error is selecting Grounding – General (EA) instead of Grounding – Other (SF). When changing components to a selection with a different Unit of Measure (UoM), make sure to recheck quantities after saving. When entering equipment, the selections that do not contain a capacity are usually out of scope. For example, Air Handling Unit – Central Station is out of scope because there are specific capacity selections available, such as Air Handling Unit – Central Station – 20,000 CFM, that should be used. A thorough understanding of the Army catalog is essential to effective QC.

A.2.3.2 Quantities

- Do quantities align with the building area? Many components are based directly on the building area such as branch wiring/grounding. Others use formulas from the BUILDER manual that incorporate building area and unit cost. It is important to make sure these quantities have been entered correctly as these are often high-cost components and may significantly skew the Annual Work Plan.
- Interior finishes should be checked to make sure the quantities are reasonable for the building's area and usage. For example, the total floor and ceiling finish quantities should never exceed the building area or the floor deck for an individual floor. In most finished buildings, the floor and ceiling finishes will be within 10 percent of the building area. However, based on usage, there could be minimal finishes, such as in warehouses.
- Look for unusually high or low quantities. This is best done by filtering UoM. Start with EA (each) and look for high quantities. Components such as lighting fixtures, fire devices and doors will often have high quantities in large buildings, but many components should not. Then filter for LF (linear feet) and SF and look for very low quantities. There are no definitive rules, but this process is a best practice for finding quantity issues.
- A common source of quantity errors is when a component was changed to a different selection with a different UoM.

A.2.3.3 Install Dates

The Year Installed dates are a critical data point for accurate condition ratings and degradation modeling in BUILDER. This data field should get a high level of QC as incorrect dates will negatively impact BUILDER's metrics and work plans.

- Newer buildings (less than 20 years old) will mostly have components that are installed the same year as the building construction date. Exceptions include some short-design-life components such as carpet.
- Year Installed is more difficult in older buildings because many components have been replaced and renovations have occurred, often multiple times. If renovation dates are available, those should be used to help date components. Also, equipment data plate information such as serial numbers can be used to verify the year a component was manufactured.
- Check the building comments for renovation dates or additional construction dates. There should be several components that match the date of the noted renovation/addition.
- Use the design life of components as a check to validate install dates. Components that are over 100% of design life and rated in good condition have suspect dates. For example, 1970s carpet that is rated as G- probably has an incorrect Year Installed date.
- Make sure the Estimated Date check box is being used correctly. **Selecting the Estimated Date box should be done with caution.** This is not a benign data point. It often has a significant effect on condition ratings. The date does not have to be exact, and for long-life components, such as piping and wiring, being within 10 years does not require the Estimated box to be checked. When the Estimated box is checked, BUILDER gives more weight to the assessor's inspection DCR. When it is not checked, more weight is given to the Year Installed date, and the component SCI will be much lower in most cases. If a component is judged to be original to the building, the Estimated Date should not be checked, even in very old buildings. Components such as foundations, superstructure, piping, and wiring are usually original, even in very old buildings, unless an addition of major renovation has occurred. See the guide below (Figure 34: Estimated Date Decision Guide) on selecting the Estimated Date check box.

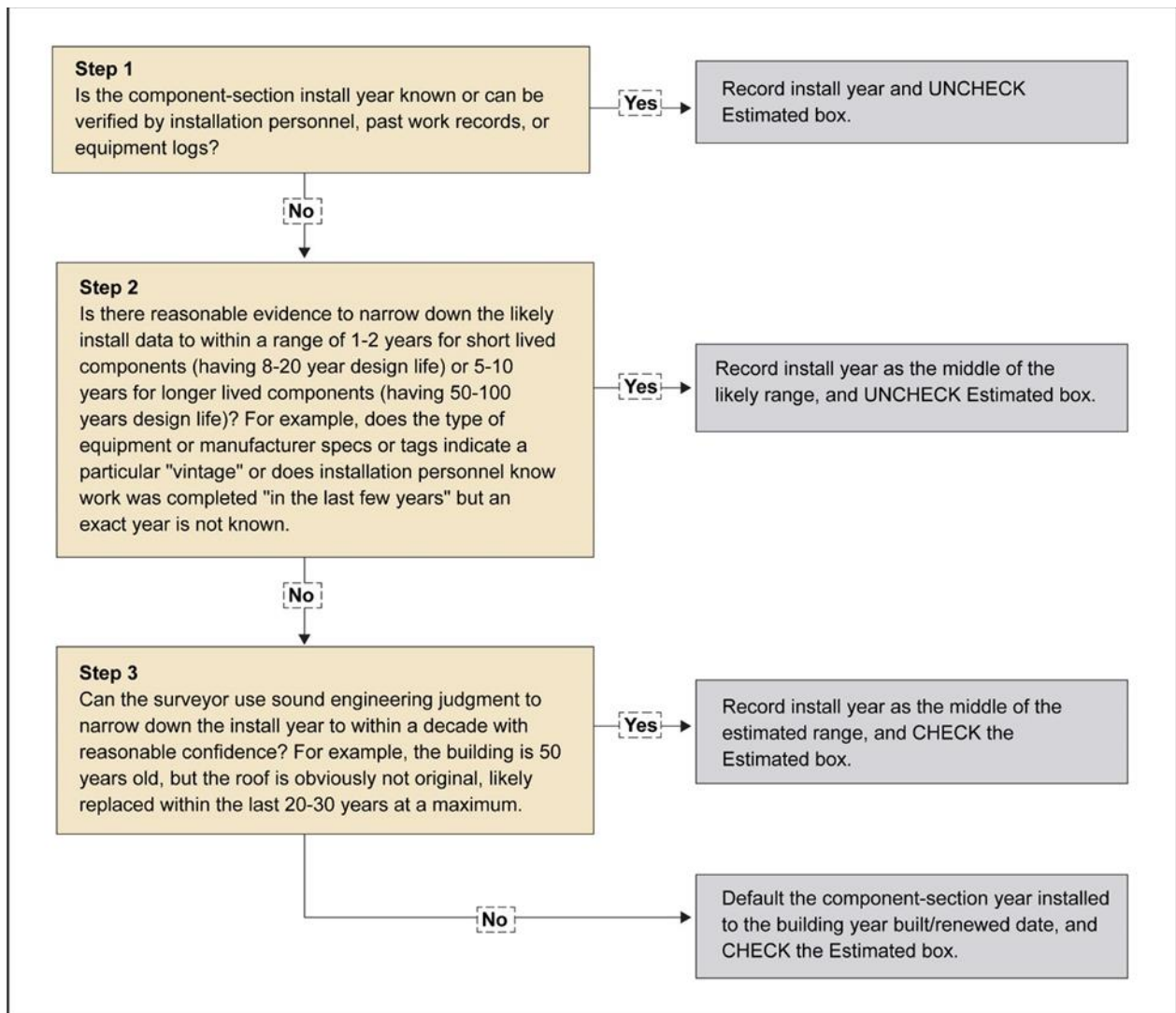


Figure 34: Estimated Date Decision Guide

A.2.3.4 Photos

The BUILDER manual provides guidance on which component sections require in inventory/section photo.

- Make sure all equipment components requiring a photo have a photo linked at the section level and not section details.
- All component selections using "Other" must have a photo.

A.2.3.5 Sectioning

The BUILDER manual has very specific rules for sectioning components. These must be followed for consistency of data. The following rules apply:

- **Floor Level** – Sectioning by floor. In multi-story buildings as well as single-story buildings with a basement, most components must be sectioned by floor. Building-wide system components such as piping, wiring, ductwork, fire alarm devices are sectioned by building and not by floor.
- **Install Date** – Different Install dates require a separate section.
- **Condition** – The manual provides guidance on sectioning by condition to prevent over sectioning.
- **Quarters/Barracks** – For certain components in Barracks/Quarters there are specific rules on sectioning. Plumbing fixtures, lighting fixtures, and interior finishes must be sectioned by quarters and commons.
- **Grouping** – Make sure grouping rules are followed as outlined in the component catalog. Most equipment components such as boilers, air handlers, chillers, and many others, must be sectioned separately (1EA) and cannot be combined into a single section.
- **Over Sectioning** – Over sectioning is a common mistake. For most non-equipment components, if they are the same component selection, install year, and floor level they should have a single section. Condition also determines when a separate section is required based on the manual rules.

A.2.3.6 Section Naming

Section Names are important to help locate the component section. Review Section Names to make sure the manual rules are followed and that section names are consistent and not over complicated.

- For multi-floor buildings the section name for interior components must include the floor level, FL1, FL2 and so on, unless the component is a building-wide system. For example, in a two-story building, plumbing fixtures will have two sections named FL1 and FL2. Domestic water piping and sanitary piping will each have a single section with no section name.
- Sectioning by room is not required for non-equipment components. For example, water closets should not be sectioned by Men's and Women's restrooms unless dates and condition require a separate section.
- Section naming for equipment components is covered in the manual and should include floor, room, and ID#. For example: FL1 RM 103 – AHU4. If there is no ID then the location will be the section name.
- Exterior section names should not include floor level. If the component section exists on a single side of the building, cardinal direction should be included. Do not include "EXTERIOR" in the section name if the component selection indicates an exterior component. For example, exterior lights on multiple sides of a building should not have a section name. However, a condensing unit will be a quantity of 1 EA and will have a section name such as WEST – CU2.
- The component type should not be repeated in the section name.
- Section names should be ALL CAPS.

A.2.3.7 Details

Dynamic equipment components will usually require details. The manual catalog will indicate which components require details and how the fields should be used. QC should ensure that:

- Details are complete and those not available are marked NA.
- Details are ALL CAPS.
- Equipment Type, Location Date manufacture and date installed should never be blank or NA.
- Capacity should not be blank except in rare cases, such as Fire Alarm Control panels.
- Capacity must contain a UoM that aligns with the component selection and includes Estimate (EST) if estimated.
- The install date in Details must match the section date.
- Control type is entered for HVAC equipment.
- ID should only contain values actually marked on the equipment.
- The capacity listed in Details aligns with the component type selection. If the exact capacity is not available as a selection the next larger component selection should be used.
- A standard section details comment is entered addressing NA entries in Details. The list of standard section detail comments are included in the manual.

A.2.3.8 Inspection Review

All visible components should have an inspection DCR (G/A/R) that aligns with the date of the assessment. Components that are not visible or mostly not visible should not have an inspection DCR. Some components assessors rarely rate include foundations, wiring, and piping. Standard comments for non-visible components are listed in the manual. The following QC checks should be performed:

- All visible components have a new DCR or a standard comment stating the component is not visible and will be rated by BUILDER's "age-based degradation."
- Components that are typically not visible do not have an inspection and have a standard comment.
- All inspections are Direct Condition Rating (DCR) and not Distress Survey.
- All components that do not have a DCR have a section inventory comment that is an exact match from the list in the manual.
- All components with a DCR of A+ or below have an inspection comment that matches the format outlined in the manual.
- All components with a DCR of A+ or below have an inspection photo.
- The design life of the component, install date, and condition should align. For example, a component with a design life of 20 years and an install date of 1980 would most likely not have a DCR of G-. The install date or rating in that case is most likely incorrect.
- The severity term used in the inspection comment aligns with the DCR.

A.2.3.9 Formatting

- While formatting does not impact cost, it does make the data look more uniform and easier to read and understand. QC should ensure that:
- Section names and details are ALL CAPS.
- Section names are consistent.
- Inventory/section comments and Inspection comments are written in complete sentences.
- Inspection comments follow the format outlined in the manual.
- Spelling is correct.
- Only the standard comments in BUILDER are used where applicable, and they are exact matches (copy/paste) of the text in the manual.
- Building comments are uniform and complete.

Appendix B. Roles and Responsibilities

Organizational roles and responsibilities from the HQ AMC, G4 level to the Installations must work in tandem to provide the BUILDER data sustainment needed for JMC planning and decision making.

B.1 HQ AMC/G4

- Provide program direction, policy guidance, and oversight to the AMC BUILDER Program.
- Coordinate with higher, adjacent, and supporting organizations to understand the status and progress being made within the BUILDER program.
- Establish an SMS Cell within the AMC/G-4 to facilitate and support the transition from ISR-I to BUILDER for sustaining inventory and facility condition data.

B.2 HQ JMC

- Provide a primary and alternate BUILDER Point of Contact for JMC HQ and each installation to the AMC BUILDER Program Manager.
- In coordination with Subordinate JMC (OIB) Installations, implement BUILDER Sustainment activities IAW this Guide.
- Perform Quality Assurance (QA) on BUILDER data to assure compliance with Army SMS Inventory and Assessment Guide, 1 Jun 19.
- Compare work order databases utilized within JMC with updates in BUILDER.
- For installations exercising command and control over Government Owned Contractor Operated (GOCO) subordinate installations, ensure the operating contractor is utilizing BUILDER. Incorporate the utilization of BUILDER and sustainment of data in future GOCO facility contracts.
- For installations exercising command and control over Government Owned Government Operated (GOGO) subordinate installations, ensure the installation is utilizing BUILDER. Identify any resource and/or manpower shortfalls via the chain of command.
- Incorporate BUILDER metrics into the Annual Work Plan prioritization process. Metrics include but are not limited to the FCI and the BCI.
- Conduct appropriate coordination with HQAMC, HQIMCOM, and ID-S IAW OPORD 20-135.
- Include BUILDER in Triannual Briefs to the AMC CG
- Validate / QA installation BUILDER Sustainment activities (via BRED files)

B.3 OIB Installations / DPWs

- Ensure appropriate and timely communication of real property inventory status and SMS BUILDER sustainment through JMC and AMC/G-4.
- Provide a primary and alternate BUILDER team lead to HQ JMC BUILDER POC.
- Ensure all real property information contained in BUILDER is RPIM compliant. Also ensure it complies with the use of GFEBS as the only approved system of record for the Army which feeds the Headquarters Installation Information System (HQIIS).
 - Organizations operating as tenants on Navy Installations are required to coordinate with the AMC BUILDER PM to top load Real Property updates from the internet Navy Facility Assets Data Store (INFADS).
- Identify any facilities that do not comply with BUILDER Sustainment standards per this Guide or that were not previously assessed in BUILDER.
- Review BUILDER SMS Annual Work Plans each FY-Q4 and validate FCI data for accuracy.
- Report annual completion status to HQ JMC NLT 15 July.

Note: These responsibilities may change over time.

Appendix C. Training Options

C.1 BRED files

According to the BUILDER SMS Guidance Manual, users can perform manual collection and data entry using the SMS BUILDER tool: BUILDER Remote Entry Database (BRED). BRED is a standalone field tool designed for use with BUILDER. It allows users to:

- Capture and collect inventory and condition assessment data using a Windows based device (tablet, laptop, etc.).
- Load inventory, assessment data, and images into SMS BUILDER database.

C.1.1 How to acquire BRED

There are several ways to obtain the BRED software.

- Downloadable software and detailed instructions on using the BRED files can be found at the SMS support documents website: <https://support.sms.erd.c.dren.mil/downloads> and at <https://support.sms.erd.c.dren.mil/downloads/bred-versions>.
- Email requests for support at: SMSSupport@erd.c.dren.mil.
- Contact the **AMC BUILDER SMS Support** desk at buildersmssupport@jacobs.com.

When performing condition assessments, assets and their subparts can be created in BRED, then imported into SMS BUILDER. The buildings (with real property information) are in SMS BUILDER and can be exported to BRED. The BRED application does not require internet access.

Inspectors can use BRED to enter inventory, equipment details and condition assessment data with the most current guidance and import the BRED file into SMS BUILDER.

C.1.2 BUILDER and BRED

Various tasks can only be done in BUILDER, and others are more efficiently done in BRED. This section highlights the differences between the two and provides guidance for when to use one over the other.

C.2 BUILDER Training

Training is a critical component to sustainment and keeping the BUILDER data relevant. Installation BUILDER POCs are strongly encouraged to take training to understand how to validate real property data, perform and update inventory and condition information, as well as how to produce, interpret, and harness the information in the reports to their advantage.

ERDC-CERL's SMS Technical Center of Expertise (TCX) offers a broad range of on-line courses that can be accessed through their Learning Management System (LMS). The SMS LMS training website is located at <https://training.sms.erd.c.dren.mil>. These courses are recommended for installation POCs to acquire basic understanding of BUILDER and the BRED software.

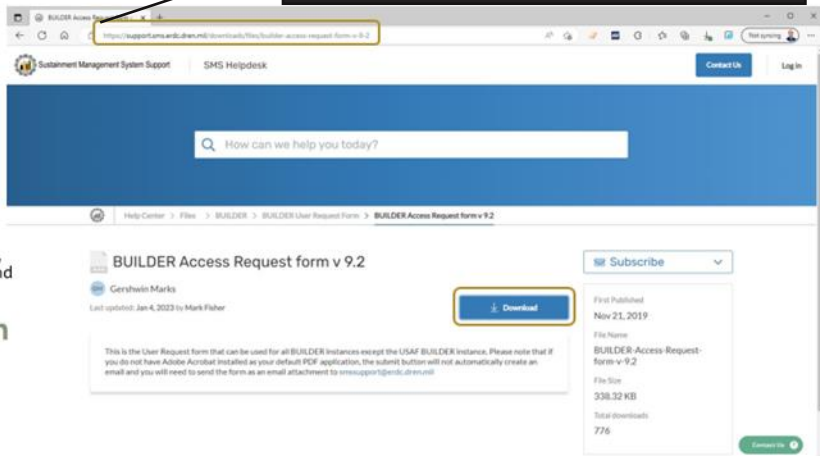
Additionally, AMC G-4 has made in-person training available to JMC and other OIB installation BUILDER POCs through a BUILDER SMS Support contract. Contact the AMC BUILDER SMS Support desk at buildersmssupport@jacobs.com for more information.

Note: SMS LMS requires a separate access request and is not the same login credentials as BUILDER SMS.

Requesting BUILDER Access

<https://support.sms.erdc.dren.mil/downloads>

- **Download BUILDER User Request Form**
 - Indicate in the comments whether training has been completed or training was received during a charette
 - For Read Only access: Complete Fundamentals and Inventory Overview courses
 - For Assessor access: Complete Fundamentals, Inventory Overview, BUILDER Indices, Work Planning and Scenarios courses
- **Send Completed Form**
 - E-mail completed form to the AMC BUILDER Support team at buildersmssupport@jacobs.com



3

Figure 35: Steps to Request BUILDER Access

To establish a BUILDER SMS account, download and complete the BUILDER Access Request Form and send it to the AMC BUILDER SMS Support desk (buildersmssupport@jacobs.com).

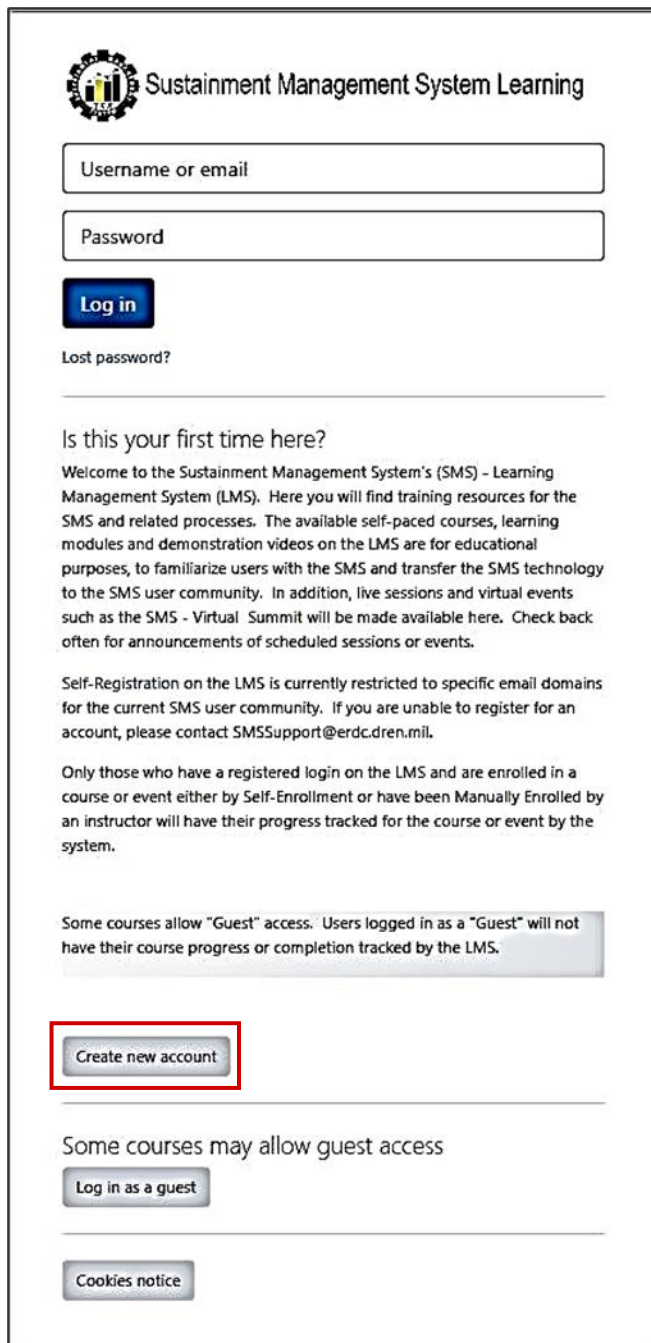
The BUILDER Access Request Form can be downloaded from the SMS website:
<https://support.sms.erdc.dren.mil/downloads>.

C.2.1 How to access SMS LMS for BUILDER Training

Although some training can be completed as a guest, it is recommended that each user create an account for a full offering of courses. The BUILDER Training site also enables the user to keep track of the training that has been completed.

C.2.1.1 Step 1

Go to <https://training.sms.erd.c.dren.mil/login/> and select **Create new account**.



 Sustainment Management System Learning

Username or email

Password

Log in

[Lost password?](#)

Is this your first time here?

Welcome to the Sustainment Management System's (SMS) - Learning Management System (LMS). Here you will find training resources for the SMS and related processes. The available self-paced courses, learning modules and demonstration videos on the LMS are for educational purposes, to familiarize users with the SMS and transfer the SMS technology to the SMS user community. In addition, live sessions and virtual events such as the SMS - Virtual Summit will be made available here. Check back often for announcements of scheduled sessions or events.

Self-Registration on the LMS is currently restricted to specific email domains for the current SMS user community. If you are unable to register for an account, please contact SMSSupport@erd.c.dren.mil.

Only those who have a registered login on the LMS and are enrolled in a course or event either by Self-Enrollment or have been Manually Enrolled by an instructor will have their progress tracked for the course or event by the system.

Some courses allow "Guest" access. Users logged in as a "Guest" will not have their course progress or completion tracked by the LMS.

Create new account

Some courses may allow guest access

Log in as a guest

Cookies notice

Figure 36: Create New Account Button

C.2.1.2 Step 2

On the New account screen, enter the requested information into the Username field, password field, Email address field, First Name field, Last Name field, City/town field, and Country field. Select the I'm not a robot checkbox, then click the Create my new account button.

New account

Username [❗]

The password must have at least 8 characters, at least 1 digit(s), at least 1 lower case letter(s), at least 1 upper case letter(s), at least 1 special character(s) such as as *, -, or #

Password [❗]

Email address [❗]

Email (again) [❗]

First name [❗]

Last name [❗]

City/town

Country

Security question [❗]

☐ I'm not a robot

 reCAPTCHA
Privacy · Terms

[❗] Required

Figure 37: New Account Screen

After the New account screen is completed, an acknowledgment email will be sent to the email address provided with a link to sign into the BUILDER Training site.

C.2.1.3 Step 3

After signing in, use the Navigation menu in the box on the left side of the SMS Learning Management Dashboard and click Site Home to display a list of courses.

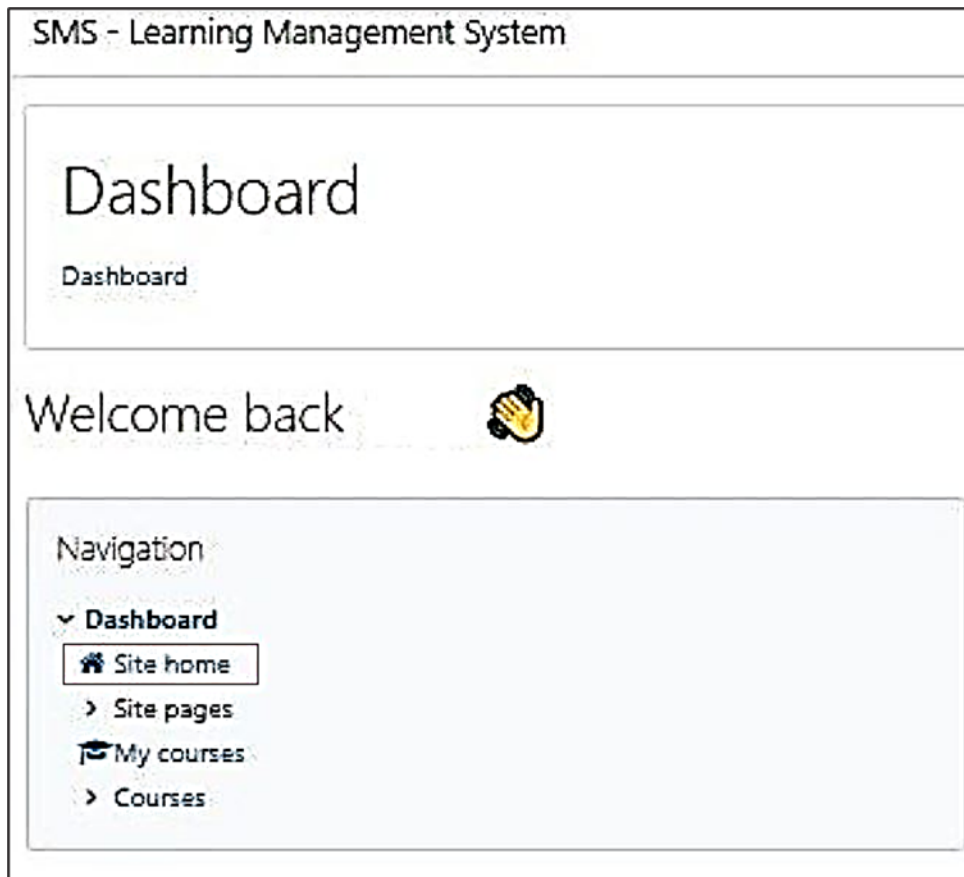


Figure 38: SMS Learning Management System Dashboard

C.2.1.4 Step 4

Navigate to the Courses area of the screen and select the arrow in the U.S. Army section to show the available courses. Click BUILDER, then select the BUILDER training.

Note: The SMS section also has a BUILDER tab, but the Army site has more detailed and specific training.

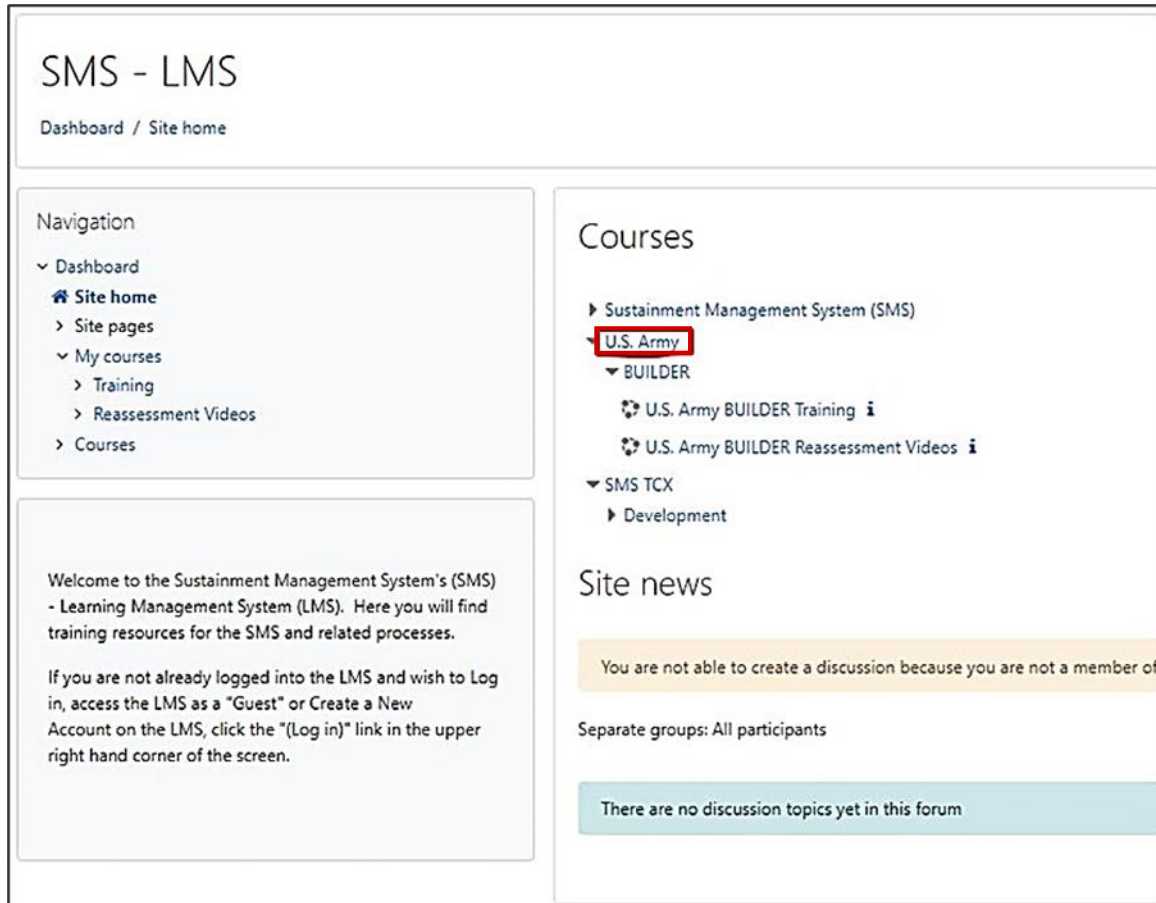


Figure 39: SMS Learning Management System Courses

C.2.1.5 Step 5

Both the U.S. Army BUILDER Training and the U.S. Army BUILDER Reassessment Videos require enrollment to access the training.

Click the blue Enroll me button for both the U.S. Army BUILDER Training and the U.S. Army BUILDER Reassessment Videos. Start by requesting the Training module before moving on to request the Reassessment Videos.

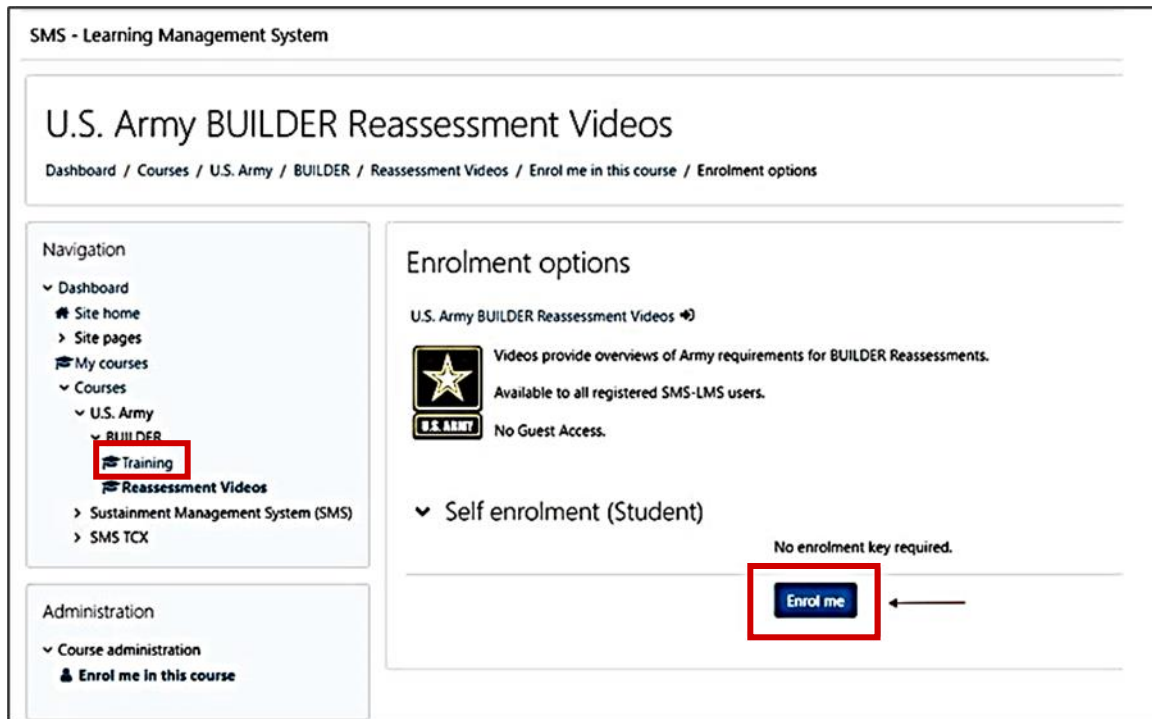


Figure 40: SMS Learning Management System Training Course Enrollment

The **Training course descriptions** explain the user access levels and recommended training for each level and covers subjects such as Fundamentals, Inventory Overview, and Data Collection. Each subject has a description, learning objective, prerequisites, and required user access level.

The screenshot displays the 'SMS - Learning Management System' interface. At the top, it says 'U.S. Army BUILDER Training' with a breadcrumb trail 'Dashboard / My courses / Training'. On the left is a navigation sidebar with sections: 'Navigation' (containing links to Dashboard, Site home, Site pages, My courses, Training, Participants, Competencies, Grades, Fundamentals, Reassessment Videos, and Courses) and 'Administration' (containing links to Course administration and Unenroll me from Training). The main content area is titled 'Course Description' and contains the following text:

Currently there are three Army Permissions Roles for user levels of access to U.S. Army BUILDER. These are **Read-Only**, **Assessor** and **Work Planner**.

A **Read-Only** (BUILDER Read-Only) permissions user will be able to pull "standard" and "custom" reports within BUILDER at or below the user's assigned Site or Organization level but not currently required by the U.S. Army for BUILDER access. This training is recommended so that a user has a basic understanding of the SMS concepts, how to navigate the

An **Assessor** (BUILDER Inspection Supervisor) will be able to inventory and access facilities in BUILDER at or below the user's assigned Site or Organization level within the Army's Database (BRED) application for field data collection.

A **Work Planner** (BUILDER Work Planner) will need to be able to run work plans, run scenarios, understand the work configuration and perform data analysis as a Facility Manager

This course covers the required training for each level of access to U.S. Army BUILDER.

Each level of access requires the completion of specific course topics as specified below. A topic is complete when all of the activities under the topic are completed.

- **Read Only:** Recommended: Fundamentals, Inventory Overview and Condition Assessment
- **Assessor:** Required: Fundamentals, Inventory Overview, Condition Assessment, Data Collection and BRED
- **Work Planner:** Required: Fundamentals, Inventory Overview, Condition Assessment, Data Collection, BUILDER Indices, Reference Books, Work Planning and Scenarios. Recommended: Fundamentals, Inventory Overview, Condition Assessment, Data Collection, BUILDER Indices, Reference Books, Work Planning and Scenarios.

Once the required topics are completed for the **Assessor** or **Work Planner** level of access a corresponding Certificate of Completion will become available for the user to print or

Course Prerequisites: None, Recommended: SMS / Overview / Introduction to SMS

To Self-Enroll in this course, select the link from the Administration menu, located on the lower left side of your screen. You must be logged into the LMS to Self-Enroll in a course.

This course is open for Self-Enrollment.

Only those who have a registered login on the LMS and are enrolled in a course by Self-Enrollment will have their progress tracked for the course by the system.

Figure 41: Training Course Description

C.2.1.6 Step 6

After the initial BUILDER Training has been completed, continue to the U.S. Army BUILDER **Reassessment Videos** for a robust offering of courses, most of which are under 10 minutes in length.

Support materials such as the full Army BUILDER SMS Inventory and Assessment Guide and BUILDER Distresses and Definitions are also included in this offering, which can be downloaded for use offline.

To access the **Reassessment Videos**, click the blue Enroll me button to access the full course of videos and downloadable documents.

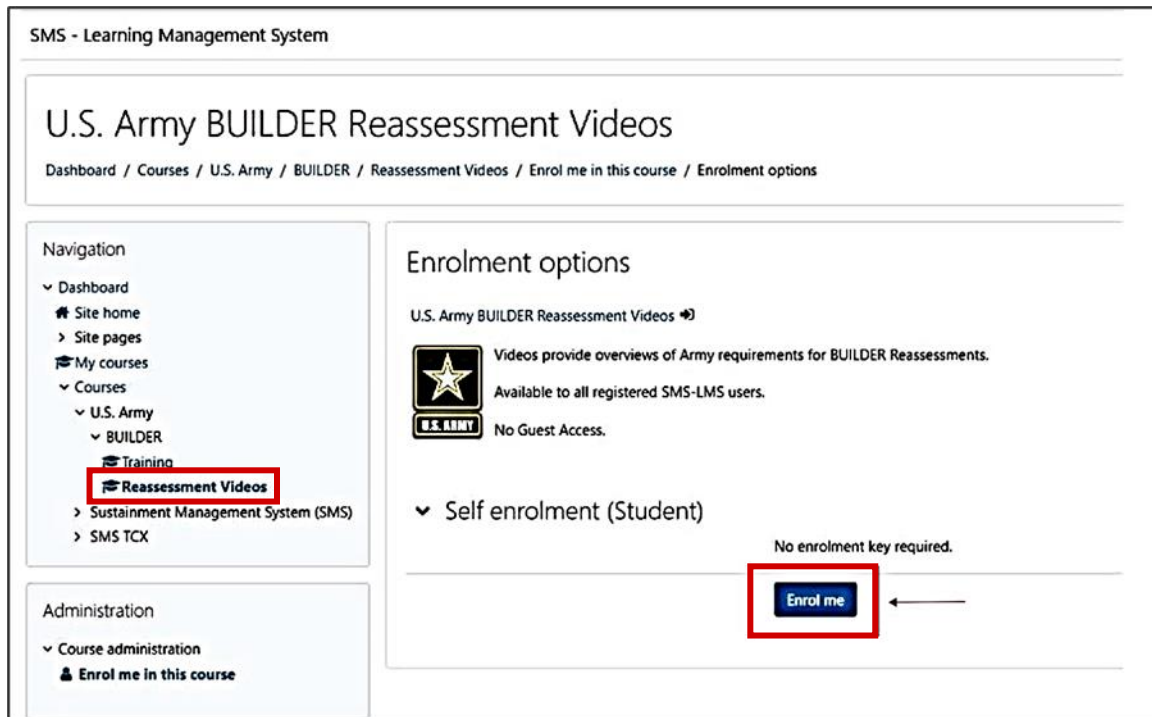


Figure 42: SMS Learning Management System Reassessment Videos Course Enrollment

C.2.1.7 Step 7

Under **Navigation**, click on any section to display the offering for that system. You can see the dropdown menu for B10 Superstructure and B20 Exterior Construction in Figure 43: Navigation.

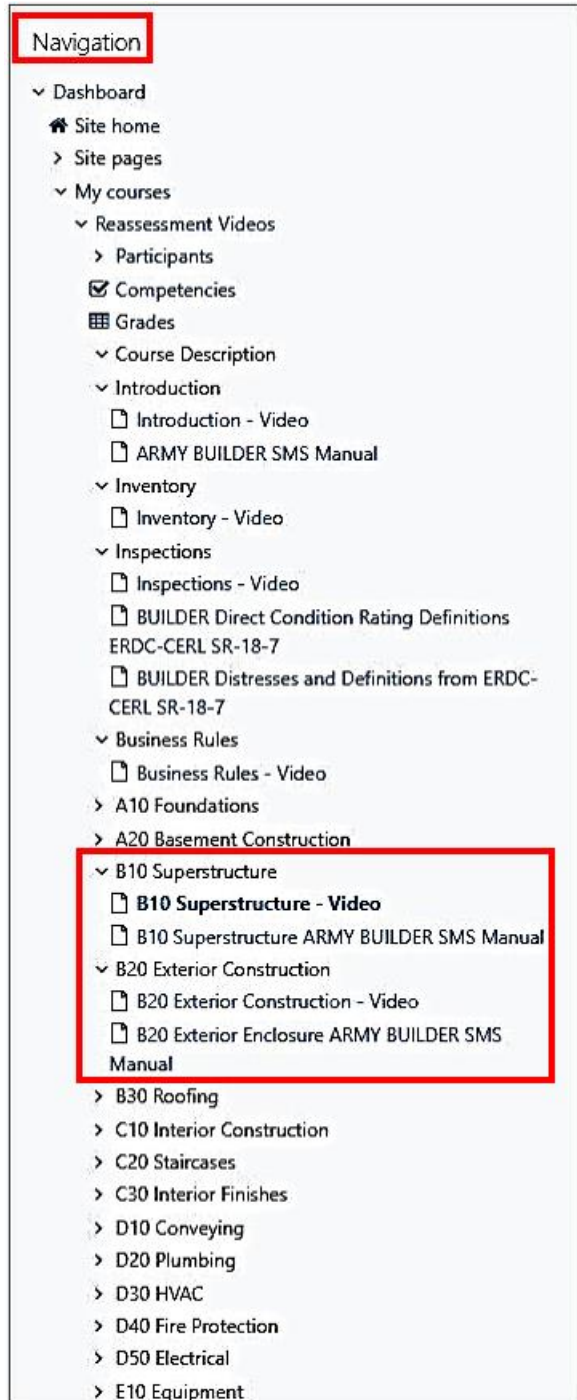


Figure 43: Navigation

C.2.2 Recommended BUILDER Training sequence

For JMC installation BUILDER POCs, the minimum recommended training courses and videos are listed below. Once enrolled (as described in Appendix C.2.1), begin with the courses in Step 1 and then view the videos in Step 2.

C.2.2.1 Step 1

Courses > U.S. Army > BUILDER > “Training” Course

- Fundamentals
- Inventory Overview
- Condition Assessment
- Data Collection
- BUILDER Remote Entry Database (BRED)

C.2.2.2 Step 2

Courses > U.S. Army > BUILDER > “Reassessment Videos” Course

- Army Reassessment Training videos:
 - Introduction
 - Inventory
 - Inspections
 - Business Rules
 - System specific videos

C.2.3 BUILDER user permissions

Table 2: BUILDER User Permissions and Associated Courses describes the user roles and associated privileges within BUILDER, along with the required or recommended courses. Most JMC Installation POCs should only need Read-Only or Inspection Supervisor (Assessor) permissions.

If a Work Planner role is required, contact the **AMC BUILDER SMS Support** desk at buildersmssupport@jacobs.com for support.

Table 2: BUILDER User Permissions and Associated Courses

Role	Privileges	Required Courses	Recommended Courses
Read-Only	View Inventory/inspection data Export Reports		Fundamentals Inventory Overview Condition Assessment
Assessor	Edit/add/delete inventory Import/export BRED files	Fundamentals Inventory Overview Condition Assessment Data Collection BRED	
Work Planner	Create or edit work plans and scenarios Create/edit scenarios and workplans Configure standards, policies, prioritization schemes, funding View/modify cost books, inflation books, service life books	Fundamentals Inventory Overview Condition Assessment Data Collection BUILDER Indices Reference Books Work Planning and Scenarios	BRED

Note: Master Planner user permission is only available to HQ Program Managers and CERL.

C.3 BUILDER Remote Entry Database (BRED)

BUILDER edits are more efficiently conducted through BRED, a standalone program that can be downloaded onto any device (laptop, tablet, etc.) and does not require connection to the internet/NIPR network. BRED looks similar to BUILDER but is a much simpler and more responsive program. All updates and BUILDER sustainment activities outlined in this Guide should be captured in BRED, then uploaded to BUILDER after Quality Control reviews are complete. (See Appendix A: BUILDER Data Quality Control Procedures for details on conducting data QC reviews.)

IMPORTANT: If photos are included in the BRED file export, they will be in a file called a bredpackage. **Both the BRED file and the bredpackage must have the same file names and be co-located to work together properly.**

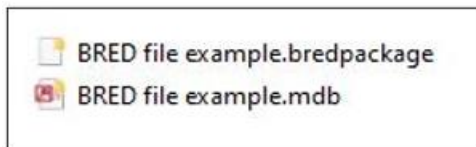


Figure 44: Example of Matching File Names for BRED and Bredpackage

C.3.1 How to download and open a BRED file

BUILDER Real Property data is top loaded from HQIIS. Once this information is in BUILDER, a BRED file can be exported at the building, complex or site level. JMC will send the BRED file and bredpackage to the installation to add or edit data. Add the files to a folder on your device.

C.3.1.1 Step 1

Download BRED onto your device from <https://support.sms.erdc.dren.mil/downloads/bred>

C.3.1.2 Step 2

Open the BRED file.

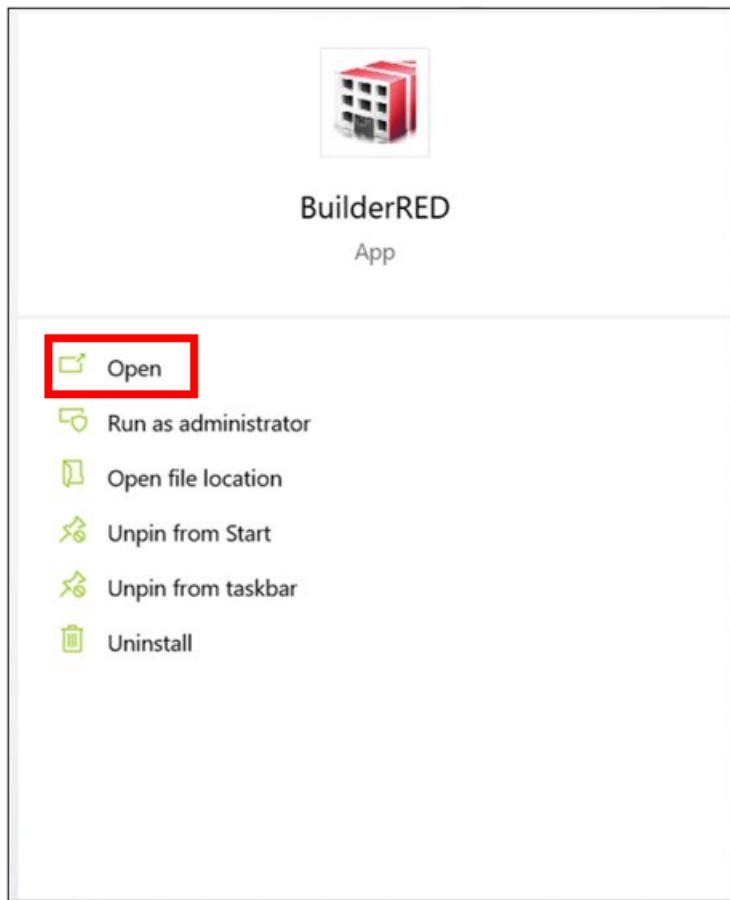


Figure 45: Open the BuilderRED File

C.3.1.3 Step 3

File > Open Inspection Data Base

Note: the bredpackage file will not appear in the folder during this step.

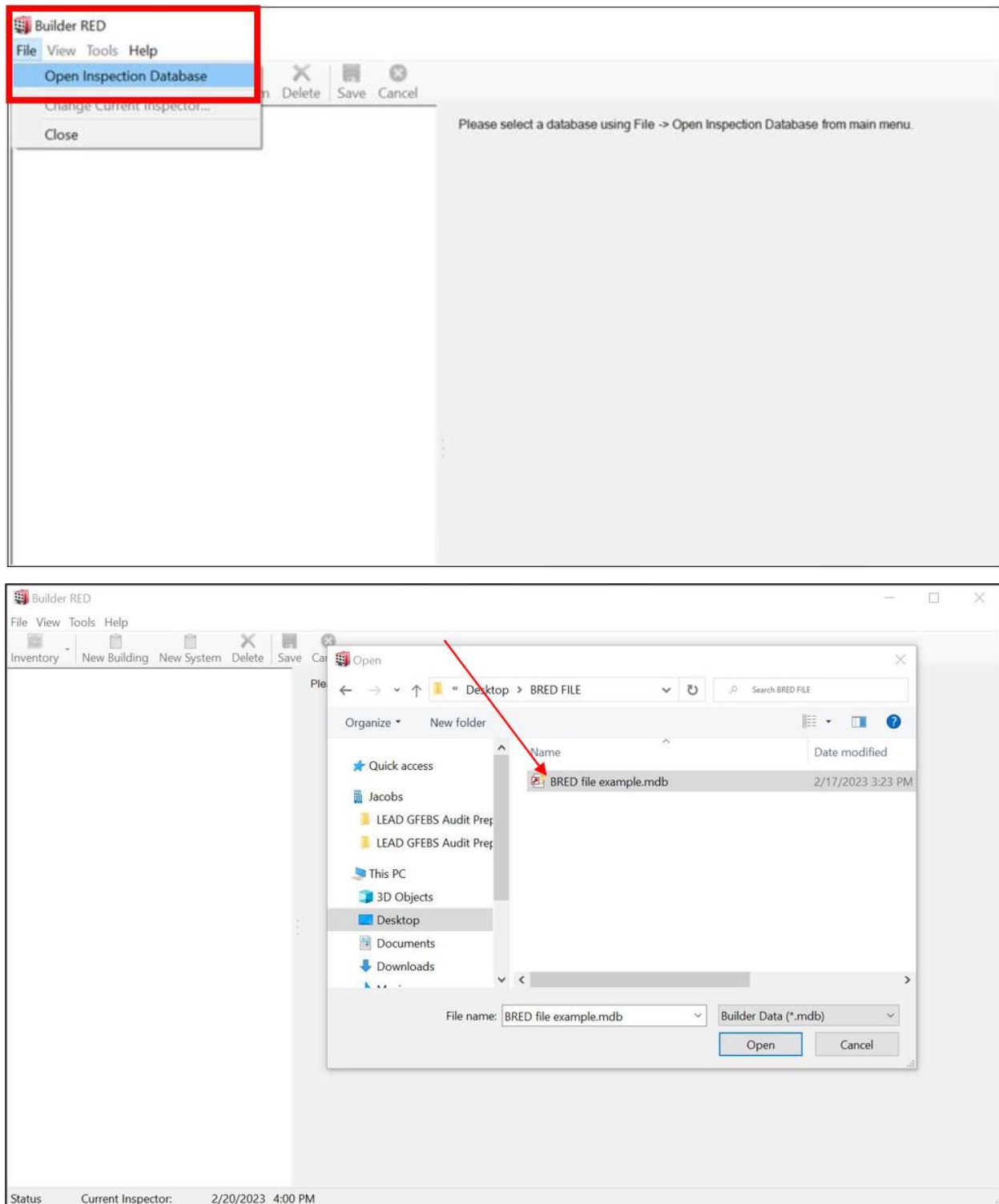


Figure 46: Open Inspection Database

C.3.1.4 Step 4

Select your name from the dropdown menu.

The screenshot shows the Builder RED application window. On the left is a list of buildings, including '32609 - HEATING PLANT'. The main area contains a form for adding a new building. The form fields are as follows:

- Building ID: 32609
- Building Use: 89121 - HEAT PLT BLDG
- Const. Type: Permanent
- Quantity: 1,600.00 SF
- Year Built: 1985
- Year Renovated: (empty)
- No. Floors: 1
- Alternate ID: 1262606
- Alternate ID Source: hqis
- Address: (empty)

A modal dialog box is open in the center of the screen with the title 'Choose the Inspector for this session'. It contains a dropdown menu with 'ALBERT, DOUGLAS' selected and an 'OK' button.

At the bottom of the window, the status bar shows: Lookup.USA, Current Inspector: 2/20/2023 4:01 PM.

Figure 47: Select Your Name from the Dropdown Menu

C.3.1.5 Step 5

Select the building to be edited from the menu and click on the + to the left of the building number to expand the tree which will show the building systems.

The building inventory data is input into BUILDER in accordance with the American Society for Testing and Materials (ASTM) International Uniformat II/Work Breakdown Structure (WBS) and contains 14 building systems.

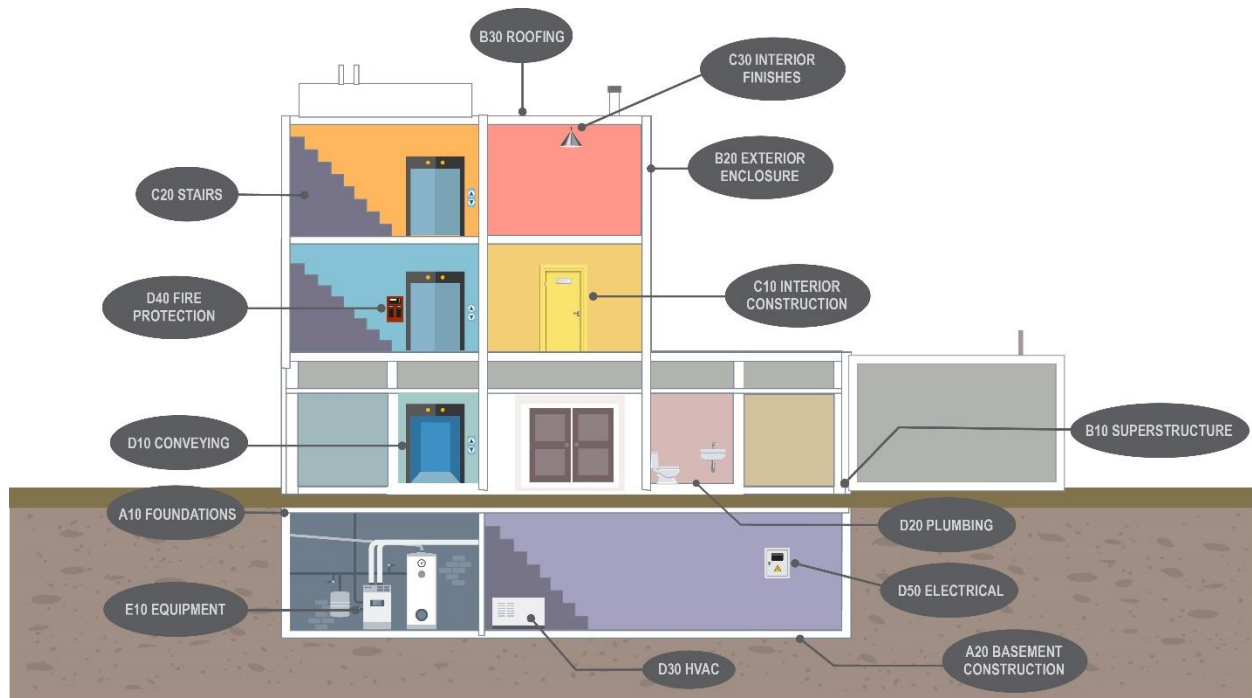


Figure 48: The 14 building systems in the ASTM International Uniformat II/WBS

C.3.2 Editing an Inventory Record in BRED

There are various factors that may require an existing component to be edited from the original record. The following illustrates some of the most common examples of when this may be required.

C.3.2.1 Editing the Component Size

1. Click on the + to the left of the component's system and find the component to be edited.

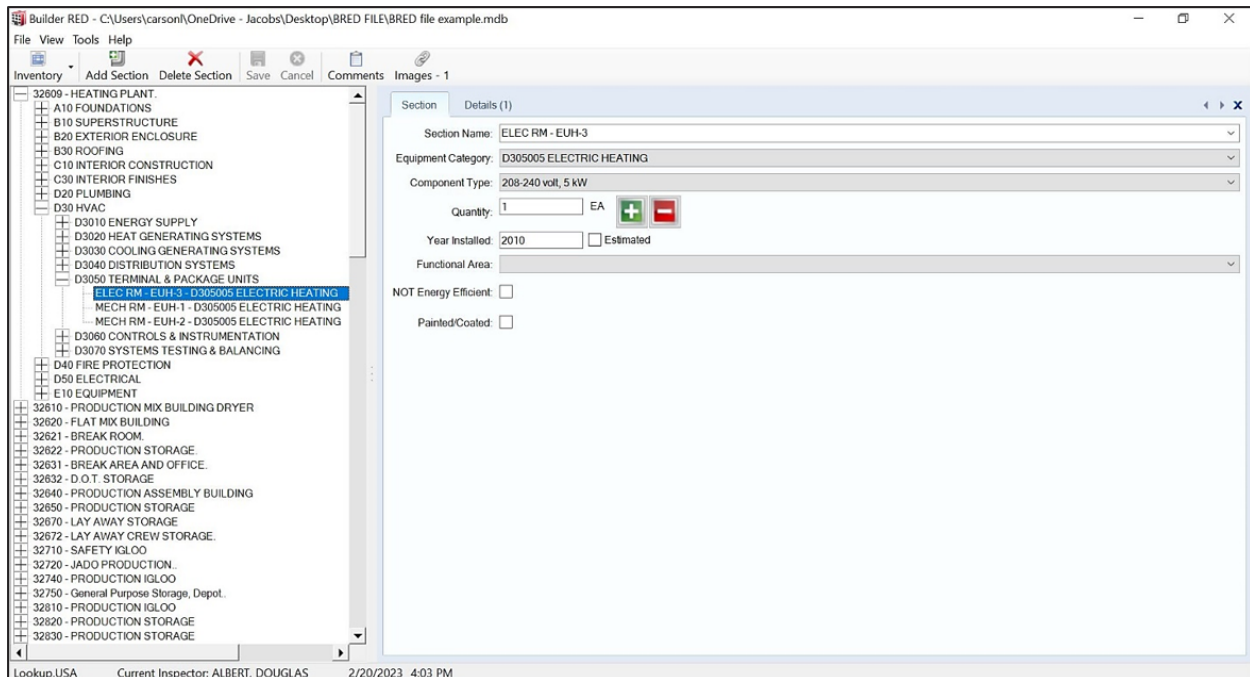


Figure 49: Editing Component Size

If section details are required (see Army BUILDER SMS Inventory and Assessment Guide, also known as “the BUILDER manual”), the Details tab will show the number of component-section details that are attached to the record. Similarly, if a bredpackage was exported and a photo is attached to the record, the Images tab at the top of the screen will indicate the number of images associated with the record. Each of these areas can be viewed by clicking on the tab icon.

Note: Both the BRED file and the bredpackage must have the same file names and be co-located to work together properly. See the example in Appendix C.3 Figure 44: Example of Matching File Names for BRED and Bredpackage

- To edit the Equipment Category and/or Component type, click on the down arrow to display the options available and select the best choice. Then click Save at the top of the screen.

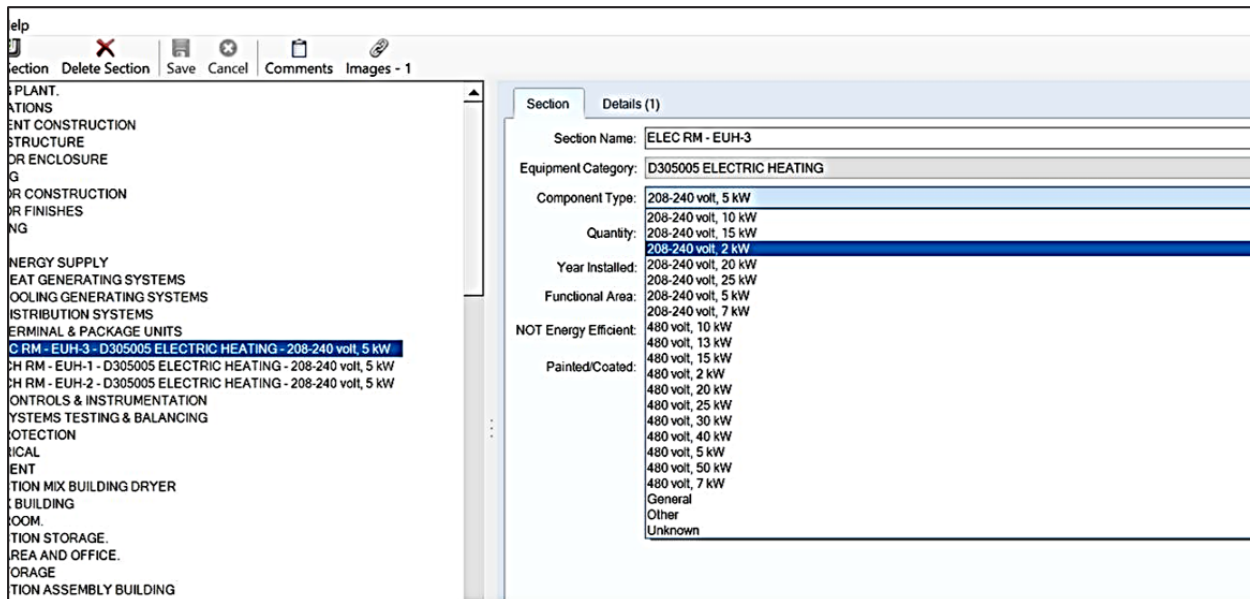


Figure 50: Edit Equipment Category and/or Component Type

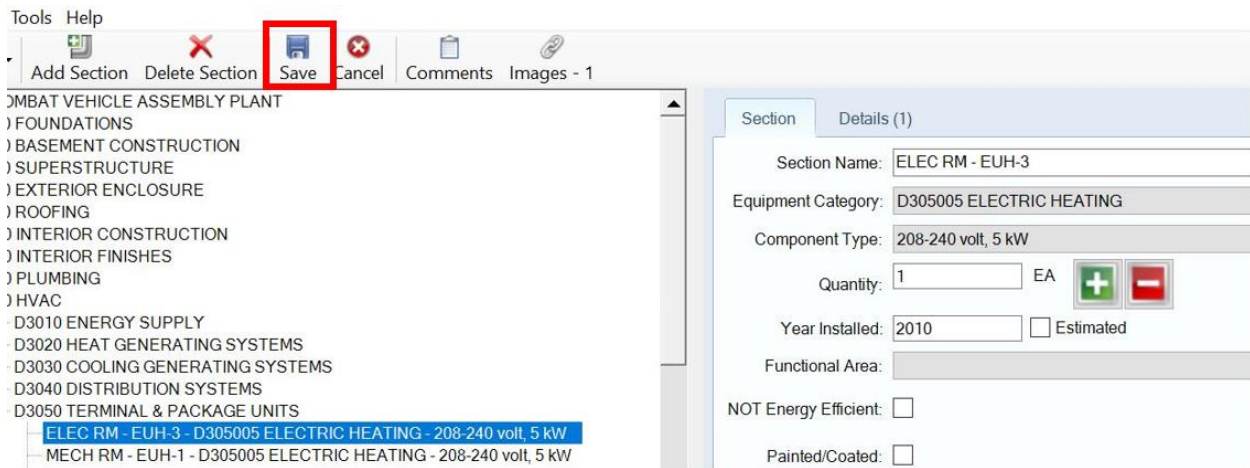


Figure 51: Save Equipment Category

C.3.2.2 Updating the Install Date

This feature is described in Figure 21: Year Installed and Figure 22: Details Tab.

C.3.2.3 Updating the Section Details

Note that if section details are required (see Army BUILDER SMS Inventory and Assessment Guide, also known as “the BUILDER manual”), the Details tab will show the number of component-section details that are attached to the record. In this example, there is one component-section details record. Refer to the following steps required to update these records as needed.

1. Open the section details record by clicking on the Details tab.

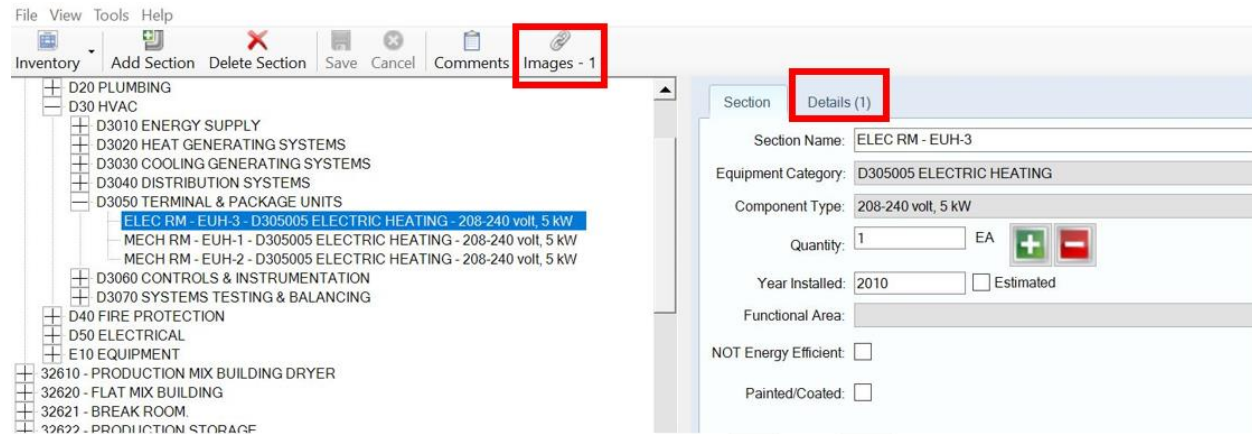


Figure 52: Section Details Record

2. For components that require section details, change the details to match the new selection by clicking on the details tab then the two blue down arrows to open the details field.

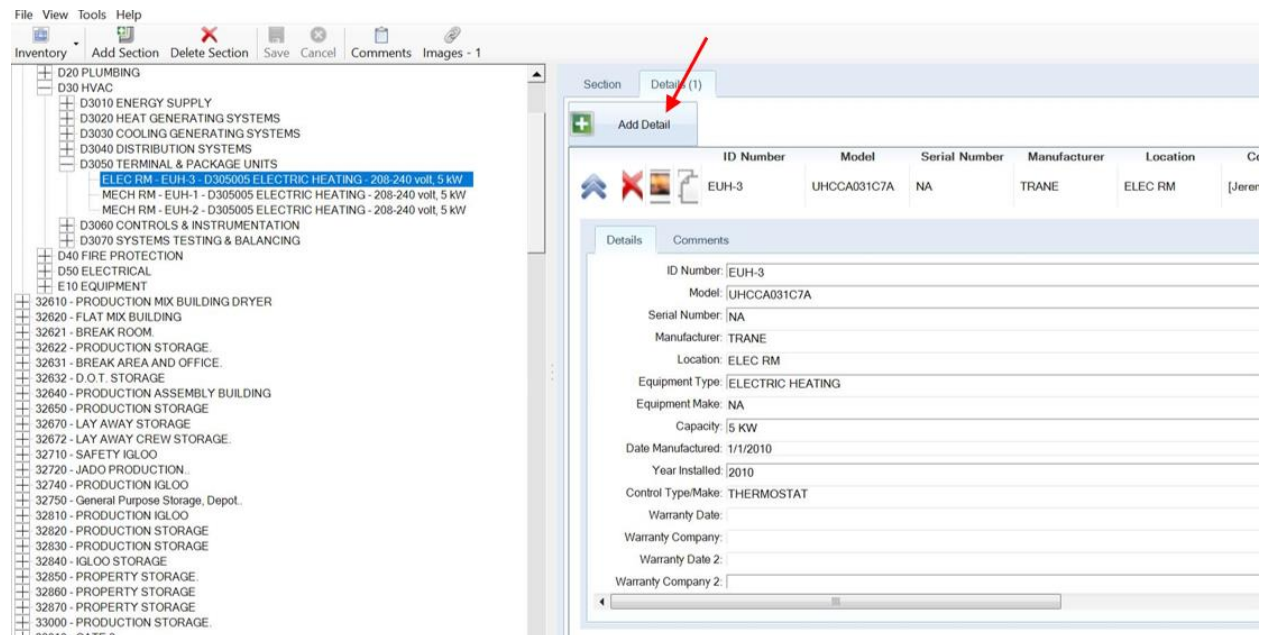


Figure 53: Add Detail

3. Change the fields as needed then click Save.

The section details comment is a standard comment when certain situations exist, such as missing nameplate information on the equipment. The various scenarios and prescribed standard comments can be found in the front of each section of the BUILDER manual.

Table 3: Standard Section Detail Comments

When	Use Standard Comment
Nameplate is not found on the component.	Nameplate not found on the component. All fields with “NA” represent data not found.
Nameplate is found on the component but it is partial/completely unreadable.	Nameplate on the component was dirty/corroded/damaged/sun washed. Section Detail fields with “NA” represent data that was not found.
Nameplate is found on the component that is readable but is missing certain Section Details fields.	Nameplate on the component was missing certain Section Detail fields. Section Detail fields have been populated and fields with “NA” represent data not found.

C.3.2.4 Updating a Section Name

1. Choose the name to be updated in BRED.

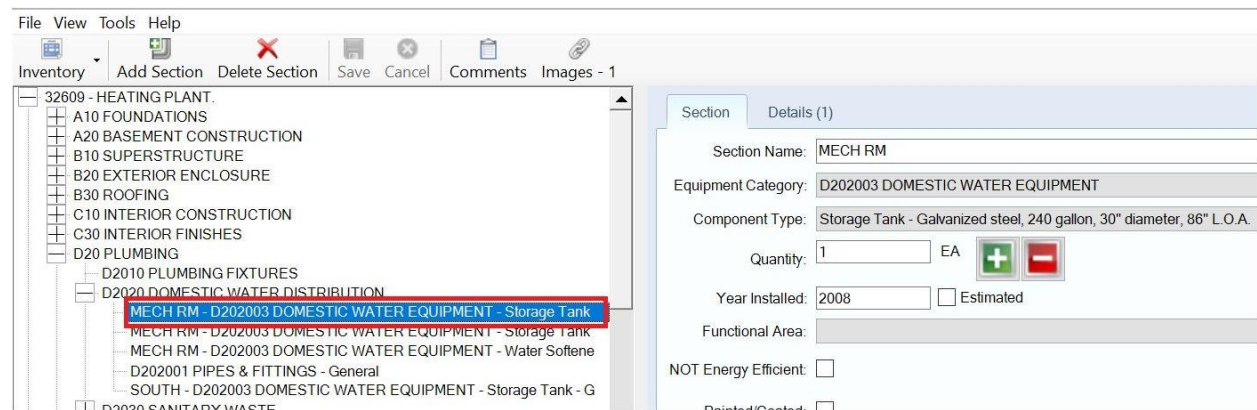


Figure 54: Find Section Name

2. Click on the section name field, change the name, then click save.

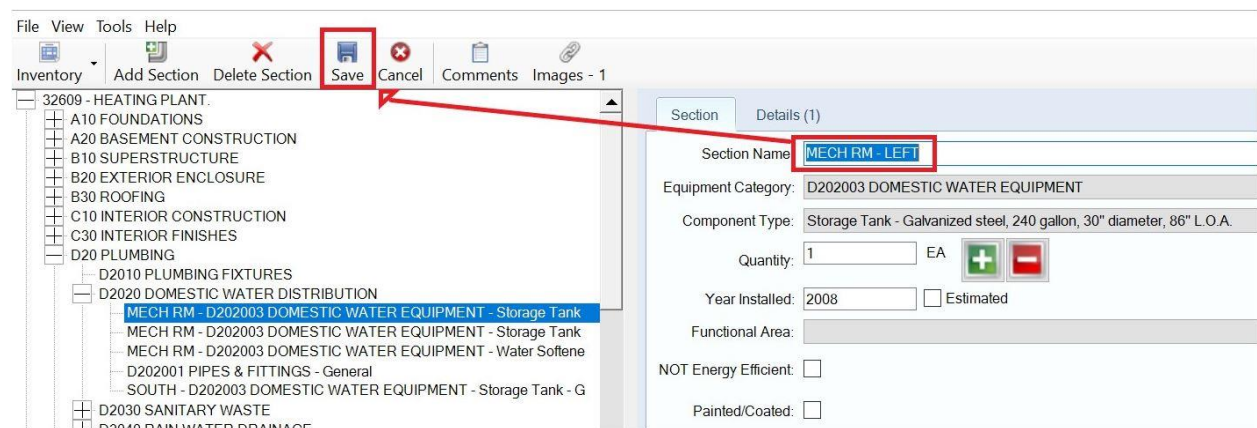


Figure 55: Edit Section Name and Save

3. If there are details for the component as indicated on the “Details” tab, the name must also be changed in Details.

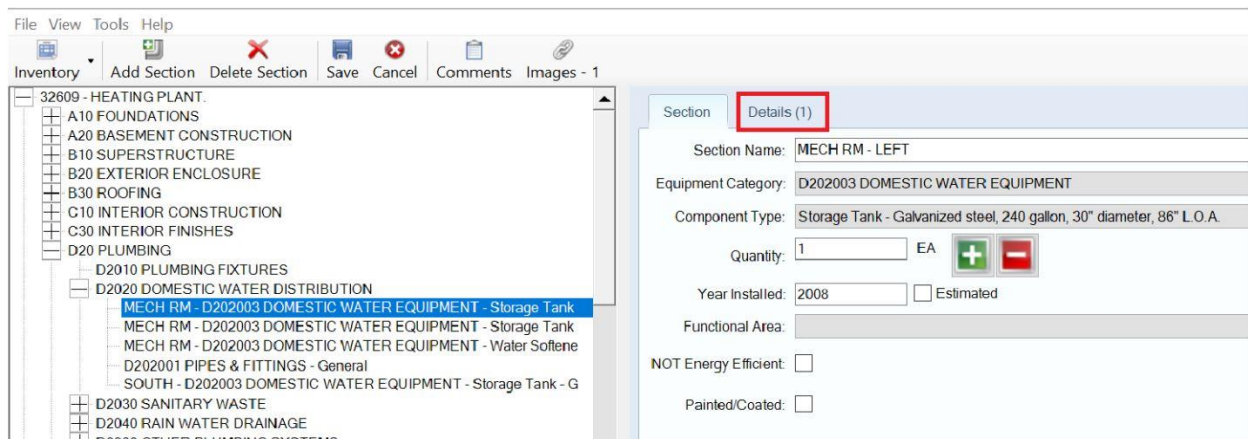


Figure 56: Section Details Record

4. Click on the details tab, then click the double blue down arrows to open.

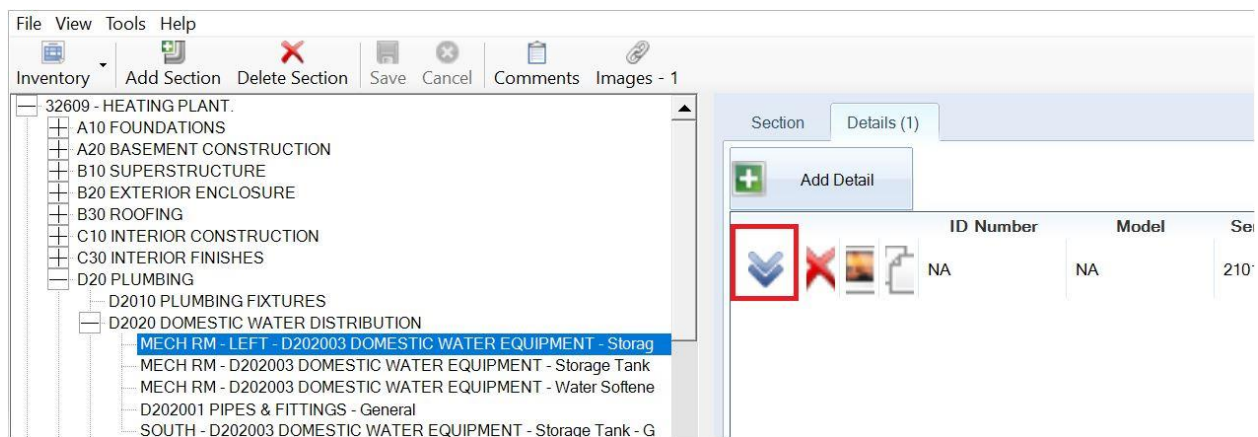


Figure 57: Opening Section Details

5. Change the name in the Location field to match the new section name, then click Save.

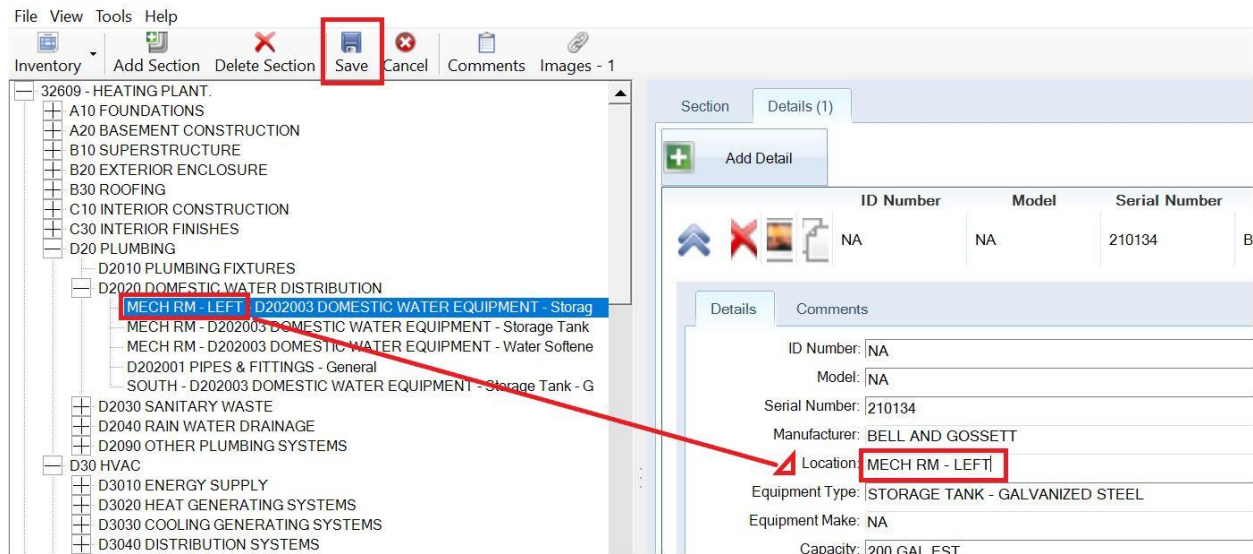


Figure 58: Add Detail and Save

C.3.2.5 Adding or Replacing a Photo

This feature is described in Section C.3.3.1 Adding a Photo in BRED.

C.3.3 Adding photos in BRED

C.3.3.1 Adding a Photo in BRED

The BUILDER manual states when an inventory photo must be attached to the component section. Photos are also required to be attached to an inspection when the rating is Amber + or lower. Therefore, some component sections will require more than one photo.

Attaching a photo to BRED is the same process for both the inventory and inspection.

1. Take a photo of the component and/or the distress with the tablet or laptop.
2. Click on **Images** on the top ribbon
3. Click Add New Image to Pending
4. Click your photo, then click **Open**. The image will be shown in the Details box. Verify it is correct.

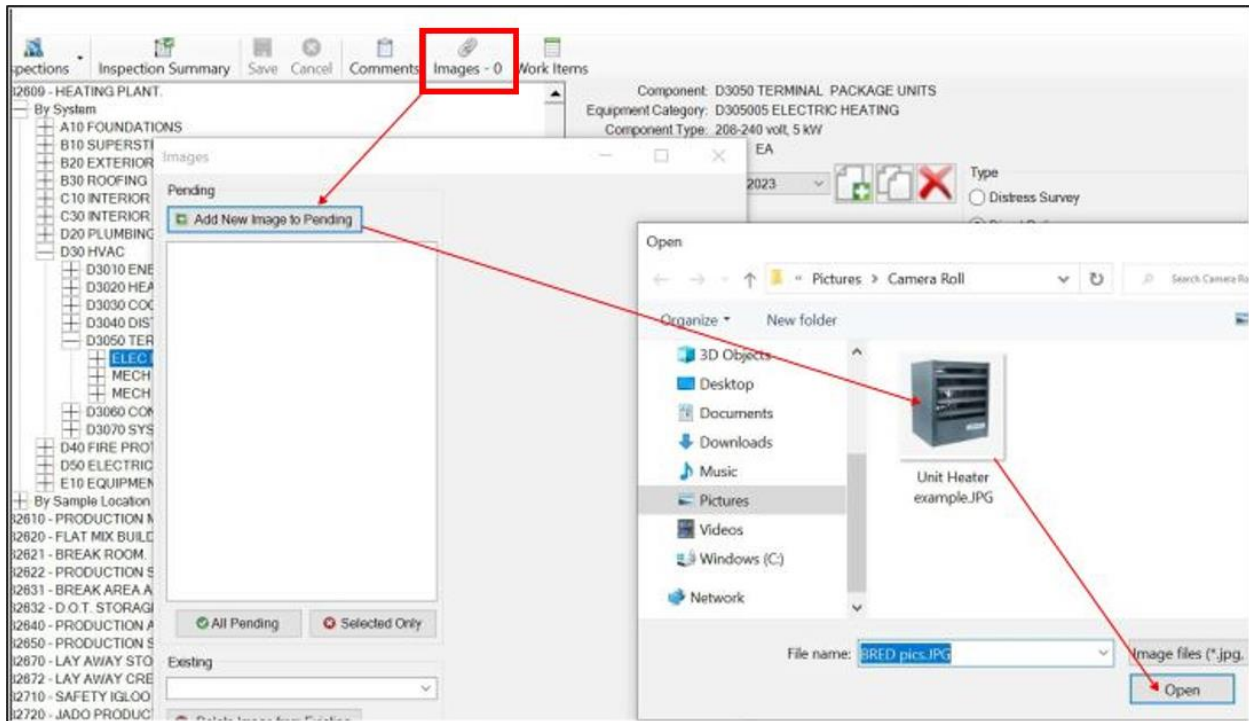


Figure 59: Attaching a Photo to BRED

5. Choose All Pending at the bottom of the window

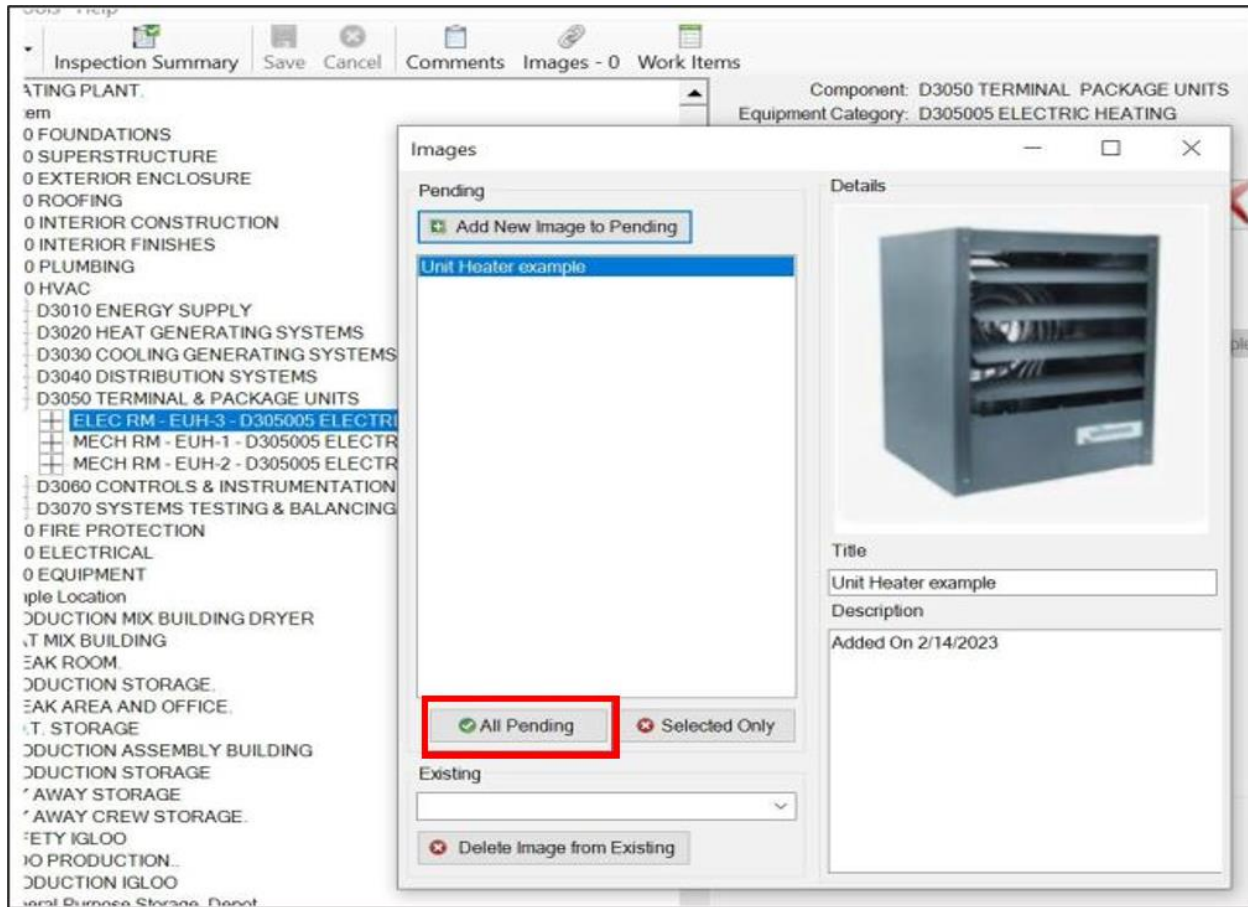


Figure 60: Add New Image to Pending

If more than one photo is needed or desired, choose all photos to be added. The list of photos will appear under the box “Add New Image to Pending”. Clicking on each of the photo descriptions will bring that photo into the box to the right for review. To remove any photo, simply click “Delete Image from Existing” at the bottom of the image box and the highlighted photo will be removed from the list.

Similarly, if a bredpackage was exported and a photo is attached to the record, “Images” at the top of the screen will indicate the number of images associated with the record. Again, in this example there is one image associated with this record.

Note: Both the BRED file and the bredpackage must be co-located and have the same file names to work together properly. See Figure 44: Example of Matching File Names for BRED and Bredpackage for an example of matching file names.

Appendix D. Most Frequently Used BUILDER Reports

BUILDER offers a wide variety of “standard” and “custom” reports that can be viewed, printed, or downloaded in multiple formats (Word, Excel, PDF). Reference USA_Custom_Report_Guide_12-08-2022 for more detail on the reports available in BUILDER.

D.1 Overview of BUILDER reports

According to SMS BUILDER Guidance, “Running a work plan is required to generate a FCI. The FCI is important as it is a financial based calculation that provides the overall health of a facility and is used to inform funding decisions. The FCI in SMS BUILDER is calculated by generating a current FY work plan and once completed, rolling up the Site (Installation or location).”

D.2 Generating BUILDER reports

1. Navigate to Site Level

Figure 61 shows the BUILDER Sustainment Management Systems interface. The left sidebar displays the 'Inventory' tree, where the site 'KS792 - USAG Camp Humphreys' is highlighted with a red box. The main panel shows the 'General Info' tab for this site, displaying various data fields and settings.

Metric	Value
CI	93
FI	100
PI	95
FCI	96

Figure 61: Generating BUILDER Reports - Navigate to Site Level

2. Select Reports in the dropdown menu

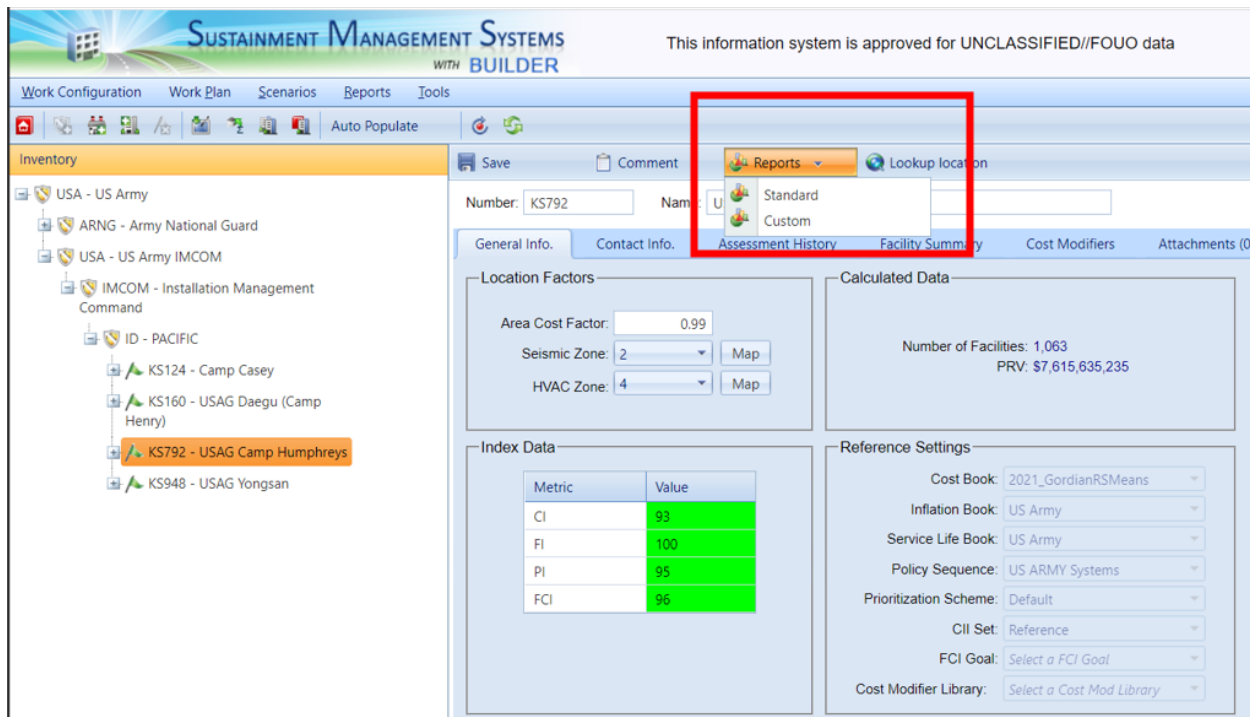


Figure 62: Generating Builder Reports - Select Reports in Dropdown Menu

3. Select desired report and then click **Run Report**.

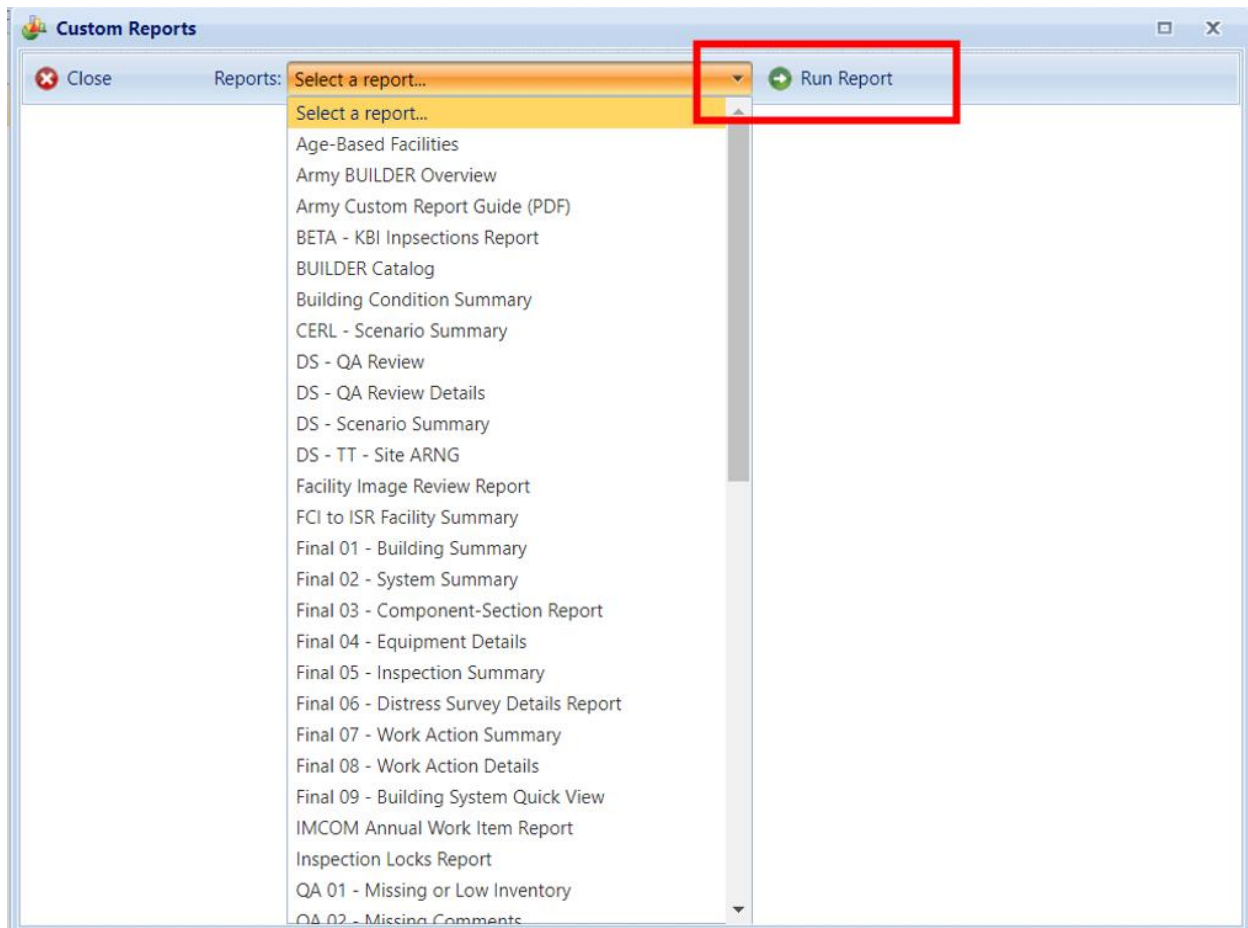


Figure 63: Generating BUILDER Reports - Run Report

4. Select Save as Excel icon to open.

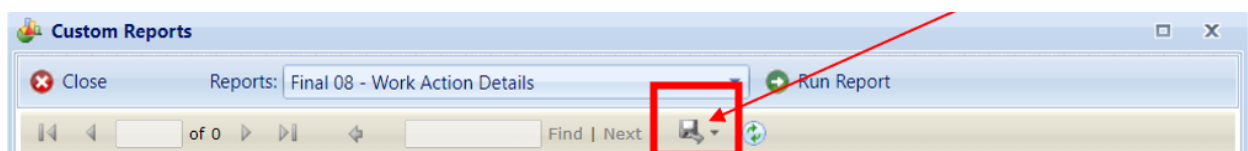


Figure 64: Generating BUILDER Reports - Save as Excel File

D.3 Most Frequently Used BUILDER Reports

The number of reports in BUILDER is varied and which report to use will depend on the information that is needed. Below is a brief description of each of the most common reports.

D.3.1 Real Property Comparison Report

Required for identifying which buildings should be included in JMC sustainment activities.

D.3.2 FCI to ISR – Facility Summary

This report includes real property information and Condition information at the Facility and System Level. This report displays the System CI for each of the 14 Systems assessed by the Army and includes basic real property information and condition information as well as the latest inspection date.

D.3.3 Final 01 – Building Summary

A list of Facilities entered into BUILDER where at least one Component is added. It also includes some basic facility-level data columns such as RPUID, year built, number of floors, MDI, BCI, Sum of Work Items, and number of components.

D.3.4 Final 02 – System Summary

A list of Systems and tier condition ratings sorted by facility. Report columns include Category Code and Description, BCI, System CI, and System CRV.

D.3.5 Final 03 – Component-Section Report

A list of Component Sections and their current condition index sorted by Facility, System, Component, and Sub-Component. This report includes basic Component information including Material/Equipment Category, Section Name, Section Install Date, RSL (years) and Inventory Section Comments.

D.3.6 Final 04 – Equipment Details

A list of Equipment sorted by Facility, System, Component, Sub-Component and Section. This report includes all section details such as model, serial number, equipment make and type, capacity, manufacturer, and location.

D.3.7 Final 05 – Inspection Summary

A list of inspected Sections sorted by Site, Complex, Facility, Systems, Component and Sub-Component. Information is included about the latest inspection on each section that has an inspection including the inspection rating, inspection comments, number of inspection images, and the RSL.

D.3.8 Final 07 – Work Action Summary

A list of Facilities, broken out by System and a sum of all work items generated against that System. Some of the report columns include estimated total cost of work items, CRV, and System CI.

D.3.9 Final 08 – Work Action Details

A detailed listing of all work items generated against this Facility, including the Project Name and Project ID. Each row of this report is a unique Work Item and includes Projected CI for FY, Work Item Description, Estimated Cost, Work Request Number, Actual Cost, and Completion Date.

D.3.10 Final 09 – Building System Quick View

For each facility, lists real property information and condition information at the Facility and System Level. This report displays the System Condition Index (SCI) for each of the 22 Systems and contains Attribute status/value information.

D.3.11 QC05i – Section Details with Inspections

A list of Sections entered into BUILDER with Section Details and the last Inspection. This report has many of the above report attributes rolled into a single very large report with more than 65 columns. Each line is a unique Component-Section and includes basic Facility information including Building Level Comments and number of images. The remaining columns are dedicated to the unique Component-Section including Quantity and UOM, Section Age, Install Date, Design Life, RSL, CI, inspection rating, and inspection comments. If the Component-Section requires Section Details, the report will also include manufacturer information, capacity, model number, and serial number.

Appendix E. BUILDER Business Rules

E.1 Purpose

This Appendix provides the specific guidance on the criteria used to determine what buildings (Real Property Asset (RPA) Type = B) will be inspected using the BUILDER Sustainment Management System (SMS). These criteria will also drive the data transferred to SMS from the Headquarters Installation Information System (HQIIS).

E.2 Endstate

BUILDER will align with the guidance from Office of Deputy Chief of Staff, G-9, to develop the Facility Condition Index (FCI) for the Real Property assets determined by the criteria identified in this document.

E.3 Common Criteria

Some data elements are common to all.

- The asset must have a valid RPUID
- The asset must be assigned to a valid Site (RPSUID)

E.4 Inspect / Business Rule

Items listed in the “Inspect” column are RPA that will get an inspection in BUILDER. Items listed in the “Exclude” column are RPA that will be rated outside of BUILDER.

Table 4: Inspect Business Rule

Code	Inspect	Exclude
RPA Type Code	B	L, S, LS
Facility Construction Type Code	PERM, SEMI	
RPA Operational Status Code	ACT, SEMI, NONF, EXCS, CARE	OUTG, SURP, CLSD, DISP, ENVH, TBA
RPA Interest Type Code	FEE, ONFG (IMCOM Only), ONST (ARNG – Federally Funded Only)	EASE, GVPV, LEASE, LESS, MHPI, ONFG (Non-IMCOM), ONFA, PDOM, PRIV, ONST (Non-ARNG)

E.5 Real Property Information Model (RPIM) Code Description

E.5.1 RPA Type Code

Table 5: RPA Type Code

Code	Description
B	Building
L	Land
S	Structure
LS	Linear Structure

E.5.2 Facility Construction Type Code

Table 6: Facility Construction Type Code

Code	Description
PERM	Permanent Facility (Expected to be used for more than 25 years).
SEMI	Semi-permanent Facility (Expected to be used up to 25 years).

E.5.3 Operational Status Code

Table 7: Operational Status Code

Code	Description
ACT	Asset is utilized for 6 months or more annually.
CARE	Asset has minimum maintenance for safety and security.
CLSD	Asset is not being used or maintained; no change in accountability requirements.
DISP	Asset has been demolished or conveyed or transferred to a non-DoD entity.
ENVH	Asset is currently not used; environmental concerns/issues are being addressed.
EXCS	Asset not required to meet the future needs and responsibilities of DoD; has been accepted by GSA for potential future federal use.
NONF	Asset cannot be used or occupied for any reason until functional capability is restored to a usable or habitable condition.
OUTG	Asset rights of use have been granted to a non-DoD entity.
SEMI	Asset is utilized less than 6 months annually (e.g. training centers).
SURP	Asset not required to meet the current or future needs and responsibilities of the Federal Government.
TBA	DoD has initiated action to acquire legal interest.

E.5.4 RPA Interest Type Code

Table 8: RPA Interest Type Code

Code	Description
EASE	A nonpossessory interest in real property granted to the U.S. Government by the owner that authorizes the right to use such asset in the specified manner.
FEE	An interest in a real property asset where a fee simple ownership interest is held by the U.S. Government, and a Military Department or WHS exercises accountability. Assets constructed by the U.S. Government in a foreign country and controlled by U.S. Government forces will be recorded as FEE.
GVPV	An interest in a real property asset held by the U.S. Government acquired by a mutually beneficial partnership agreement between a Military Department or WHS and a private entity, where equity interest in a project is shared for a specific business purpose. This interest type applies when the DoD has ongoing reported financial statement costs directly associated with an asset(s) gained by the project or the asset is location on a military installation.
LEAS	A possessory interest in real property conveyed to the U.S. Government for a limited period of time by contractual agreement from another party. The agreement is subject to specified conditions and the other party retains ownership of the asset.
LESS	An interest in a real property asset held by the DoD that is less than a fee interest and does not fall into any other category. The specific interest granted to the DoD must be detailed in the transfer instrument granting DoD's interest.
MHPI	An interest in a real property asset held by a private entity for military housing (family or unaccompanied) and associated assets pursuant to Title 10 United States Code, Sections 2871 to 2885.
ONFA	An interest in a real property asset where a fee simple ownership interest is held by the U.S. Government and accountability is exercised by a non-DoD federal agency. This interest type does not include public lands or interest in land owned by the United States administered by the Secretary of the Interior through the Bureau of Land Management which are to be reported as PDOM.
ONFG	A possessory interest in real property conveyed to the U.S. Government by a foreign government / host nation that constructed and holds title to the real property asset. A Military Department or WHS exercises accountability on behalf of the U.S. Government. The U.S. Government has been granted control of the asset by the foreign government/host nation by consent or formal agreement.
ONST	Real property owned or otherwise managed by a state or one of its political subdivisions in direct support of a DoD mission and DoD has shared use or sustainment responsibilities.
PDOM	An interest in public lands or interest in land owned by the U.S. Government, administered by the Secretary of the Interior through the Bureau of Land Management, where a Military Department or WHS exercises accountability on behalf of the U.S. Government.
PRIV	An interest in a real property facility that is owned and controlled by a private entity and operating with permission on U.S. Government owned land. This would include assets gained by an agreement between a Military Department or WHS and a private entity, where an equity interest in a project is shared for a specific business purpose where the DoD does not have ongoing reported financial statement costs directly associated with an asset(s) gained by the project.

Acronyms and Abbreviations

AMC	Army Materiel Command
ASTM	American Society for Testing and Materials
BCI	Building Condition Index
BIM	Building Information Models
BRED	BUILDER Remote Entry Database
CERL	Construction Engineering Research Laboratory
CI	Condition Index
CRV	Component Replacement Value
DCR	Direct Condition Rating
DCS	Deputy Chief of Staff
DoD	Department of Defense
EA	Each
ERDC	Engineer Research and Development Center (U.S. Army)
EST	Estimate
FCI	Facility Condition Index
FIP	Facility Investment Planning
FY	Fiscal year
GFEBs	General Fund Enterprise Business System
GOCO	Government Owned Contractor Operated
GOGO	Government Owned Government Operated
HAZMAT	Hazardous Materials
HQIIS	Headquarters Installation Information System
IAW	In Accordance With
iNFADS	internet Navy Facility Assets Data Store
JMC	Joint Munitions Command
KBI	Knowledge-Based Inspection
LF	Linear feet
LMS	Learning Management System
OIB	Organic Industrial Base
OSD	Office of the Secretary of Defense
POC	Points of Contact

PRV	Plant Replacement Value
QA	Quality Assurance
QC	Quality Control
RPAO	Real Property Accountability Officer
RPIM	Real Property Information Model
RPUID	Real Property Unique Identifier
SCI	System Condition Index
SCIF	Sensitive Compartmented Information Facility
SF	Square Feet
SMS	Sustainment Management System
SRM	Sustainment, Restoration and Modernization
TCX	Technical Center of Expertise (ERDC-CERL)
UoM	Unit of Measure
WBS	Work Breakdown Structure